Together First, Apart Later? When Optimizers Start to Learn Different Functions

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Abstract

From the same initialization and the same sequence of mini-batches, we train small transformer language models with SGD with momentum, AdamW and Muon, and ask when their learned functions start to differ. We measure the distance between the next-token predictions of two runs on held-out text, compare runs at equal training loss so that speed is not mistaken for difference, and calibrate against the distance between two runs of the same optimizer that differ only in data order. For each pair of optimizers we look for a persistent split, a loss level from which their distance stays above this floor, with a bootstrap over data orders, and test what coincides with the changes in distance: sharpness reaching its stability threshold, the end of warmup, or a loss level. In all three language-model settings and for every pair, the loss-matched distance is 30 to 200 times the floor at high loss, falls back to near the floor around the stage where the models pass through the bigram statistics of the data, and stays there: the only persistent split is between Muon and SGD on Shakespeare, where SGD barely trains, and it does not survive a change of metric or threshold. At equal steps the optimizers do differ, mostly because they progress at different speeds; at the end of TinyStories-30M, replacing AdamW by Muon changes the predictions no more than changing the data order. AdamW sits at its adaptive edge of stability for much of training whether or not the optimizers differ, and SGD reaches its threshold only around the end of warmup. On CIFAR-10 we replicate the break-even effects of Jastrzebski et al. (2020) on sharpness and gradient noise, but at equal loss two learning rates differ by only 1.0 to 1.75 times the floor. Swaps of optimizer at nine times during training are in progress.

1 Introduction

The optimizer decides which of the many solutions with equal training loss a network converges to. In linear models this choice is understood: a first-order method acts as steepest descent in some norm and converges in direction to the max-margin solution in that norm (Gunasekar et al., 2018), which is ℓ_2 for gradient descent (Soudry et al., 2018), ℓ_∞ for Adam (Zhang et al., 2024a; Xie and Li, 2024) and the spectral norm for Muon (Fan et al., 2025). These are statements about the limit of infinite training, approached slowly. In deep networks the comparison has been made at the end of training: adaptive methods were found to generalize worse (Wilson et al., 2017), then to learn richer features (Vasudeva et al., 2025); models with the same error can compute different functions (Cortesi et al., 2026); and on language models most optimizers reach a similar loss (Zhao et al., 2025).

We ask *when* the difference between optimizers appears. From the same initialization and the same data order, do SGD with momentum, AdamW and Muon first follow the same path in function space and only later separate? If so, where is the split, what happens in the optimization at that moment, and is the final difference already decided there?

The timing matters for two reasons. First, it says which theory is relevant. If optimizers separate early, the asymptotic margin results do not describe how the difference arises, and early events in training are candidates instead. In vision, Jastrzebski et al. (2020) show that SGD runs with different learning rates separate in prediction space early in training, and tie this to a *break-even point*; and Frankle et al. (2020) find that training becomes stable to the noise of SGD early, after which copies of a network trained with different data orders stay in one linearly connected region. Edge-of-stability events (Cohen et al., 2021; Cohen et al., 2022; Andreyev and Beneventano, 2024) are the natural candidate for such an early event, and their thresholds differ by optimizer. Second, it matters in practice. Switching optimizers during training (Keskar and Socher, 2017) assumes the second optimizer can still change the solution; a split point bounds when that is possible.

Approach. We train nanoGPT models (Karpathy, 2022) on character-level Shakespeare (10.8M parameters) and on TinyStories (14M and 30M parameters), with SGD with momentum, AdamW, and Muon (Jordan et al., 2024), each at a learning rate chosen by a sweep, from a shared initialization, with five data orders per optimizer. The distance between two runs is the Jensen–Shannon divergence and the top-1 disagreement of their next-token predictions on fixed

held-out contexts. We compare runs at equal steps and at equal training loss, and calibrate against a noise floor: the distance between two runs of the same optimizer with different data orders. A SimpleCNN on CIFAR-10 replicates the break-even analysis of Jastrzebski et al. (2020) as a check of the pipeline.

Contributions.

1. **Divergence curves** between every pair of optimizers in three settings, at equal steps and at equal training loss, against a data-order noise floor (Section 5.2). In all 9 pairs the loss-matched distance is far above the floor early and returns to near it by the bigram stage of training.
2. **A test for a split point**: episodes in which the distance exceeds the floor and a persistent split, located as a loss level with a bootstrap over data orders (Sections 4.2 and 5.2). There is a persistent split in 1 of 9 pairs (Muon–SGD on Shakespeare, from loss 2.82, in 96.5% of resamples), none between AdamW and Muon.
3. **Optimizer against data order** at equal steps (Section 5.3): at the end of TinyStories-30M the AdamW–Muon distance equals the data-order floor, and the n-gram references show where the optimizers' routes meet.
4. **What coincides with the changes in distance**: sharpness against its optimizer-specific stability threshold, batch sharpness, the end of warmup (Section 5.4). The edge of stability does not explain them: AdamW is at its threshold for long stretches whether or not the optimizers differ, and SGD only around the end of warmup. Muon with its auxiliary AdamW matched to AdamW still memorizes on Shakespeare, so the memorization comes from its updates of the hidden matrices.
5. **A causal test by swapping** optimizers at nine fork times with warm optimizer states, scored by where the fork lands between the two pure runs, and a basin-level test by linear mode connectivity (Section 5.5). [TODO: results pending: r for forks before and after the end of the early episode.]
6. **The anatomy of the sharp direction**: which parameter blocks carry the top Hessian eigenvector and its curvature, by optimizer and over training (Section 5.6). It differs by optimizer (embeddings for Muon on Shakespeare, value–output matrices for Muon on TinyStories, head or norms for AdamW), and most of the eigenvalue comes from curvature between blocks.
7. **A replication of the break-even point** on CIFAR-10 and its transfer to the language models (Section 5.7). The sharpness and gradient-noise effects replicate, but at equal training loss two learning rates differ by only 1.0–1.75 times the floor.

Code: `https://github.com/qqgjyx/MIT-6.7910` (`code/`).

2 Related work

Implicit bias of optimizers. For linear models the limit of training depends on the geometry of the optimizer: steepest descent in a norm converges to the max-margin solution in that norm (Gunasekar et al., 2018), giving ℓ_2 for gradient descent (Soudry et al., 2018), ℓ_∞ for Adam (Zhang et al., 2024a), an ℓ_∞-constrained KKT point for AdamW with weight decay (Xie and Li, 2024), and the spectral norm for Muon (Fan et al., 2025). Away from the limit, finite step sizes act as implicit regularizers: for SGD the modified loss penalizes the gradient norm (Smith et al., 2021), while for Adam the corresponding term can have the opposite sign (Cattaneo et al., 2024). Empirical comparisons in deep networks are made at the end of training (Wilson et al., 2017; Vasudeva et al., 2025; Zhao et al., 2025; Cortesi et al., 2026). We look at the whole trajectory instead.

Edge of stability. With full-batch gradient descent, the top Hessian eigenvalue rises until it reaches $2/\eta$ and then stays near it while the loss keeps decreasing (Cohen et al., 2021); this is explained by self-stabilization (Damian et al., 2023) and described by central flows (Cohen et al., 2025). Adaptive methods reach an analogous *adaptive edge of stability*, where the top eigenvalue of the preconditioned Hessian sits at $(2 + 2\beta_1)/((1 - \beta_1)\eta)$ (Cohen et al., 2022). For mini-batch SGD, Andreyev and Beneventano (2024) show that the quantity that stabilizes at $2/\eta$ is the *batch sharpness*, the curvature of mini-batch losses along their own gradients, while $\lambda_{\max}$ stays below; Andreyev et al. (2026) study the same regime with momentum. Kalra and Barkeshli (2024) relate learning-rate warmup to these sharpness dynamics. Transformer Hessians differ strongly across parameter blocks (Zhang et al., 2024b), which motivates our block decomposition.

Break-even point and early training. Jastrzebski et al. (2020) argue that early in training SGD reaches a break-even point, after which the learning rate and batch size set the curvature and the gradient noise along the rest of the trajectory, and show that SGD runs with different learning rates separate early in prediction space. Frankle et al. (2020) find that training becomes stable to SGD noise early: two copies trained on from the same checkpoint with different data orders end in one linearly connected region once the checkpoint is past an early point. We use both as templates, comparing optimizers instead of learning rates or data orders.

Table 1: Settings. L layers, model width d, vocabulary V. Parameters in parentheses exclude the token and position embeddings and the head. LM batches are sequences × tokens. "aux.": learning rate of Muon's auxiliary AdamW. CIFAR-10 settings in Section 4.9. LM schedules: linear warmup for w steps, then cosine decay to 10% of the peak learning rate at the last step. The CIFAR-10 runs use a constant learning rate.

Setting	Model (L, heads, d, V)	Params	Steps	Batch	w	η_{SGD}	η_{AdamW}	η_{Muon} (aux.)
Shakespeare-char	6, 6, 384, 65	10.8M (10.6M)	1,500	64×256	100	0.05	$3 \cdot 10^{-4}$	0.04 (10^{-3})
TinyStories-10M	6, 6, 384, 4096	13.9M (10.6M)	10,000	128×256	200	0.03	$2 \cdot 10^{-3}$	0.01 ($2 \cdot 10^{-3}$)
TinyStories-30M	8, 8, 512, 4096	29.5M (25.2M)	20,000	128×256	400	0.1	$2 \cdot 10^{-3}$	0.01 ($1.5 \cdot 10^{-3}$)
CIFAR-10	SimpleCNN, ReLU	0.36M	39,000	128	–	plain SGD, $\eta \in \{0.001, 0.01, 0.1\}$		

Muon and optimizer switching. Muon (Jordan et al., 2024) updates each hidden weight matrix with its momentum after approximate orthogonalization by a Newton–Schulz iteration, and has been scaled to large language models (Liu et al., 2025). SWATS (Keskar and Socher, 2017) switches from Adam to SGD during training, scored by accuracy; our swaps are scored by where the function ends.

3 Setup

Models and data. Table 1 lists the four settings. The language models are nanoGPT (Karpathy, 2022) with pre-LayerNorm blocks, GELU MLPs of width $4d$, no biases, no dropout, an untied output head, and separate query, key and value projections (so that Muon orthogonalizes each and the block decomposition of Section 4.7 can separate the query-key from the value-output matrices). The context is 256 tokens. Character-level Shakespeare uses nanoGPT's 90/10 split and a 65-character vocabulary. TinyStories (Eldan and Li, 2023) (GPT-4 version) uses a 4096-token byte-level BPE trained on the training split, so that the head and the stored predictions stay small. The vision check is the SimpleCNN of Jastrzebski et al. (2020) (two 32-filter and two 64-filter 3×3 convolutions with max pooling after each pair, then a 128-unit dense layer, ReLU) on CIFAR-10 (Krizhevsky, 2009) without augmentation.

Shakespeare runs take 24.6M tokens (about 24 passes over the 1M training characters). Without dropout the validation loss is lowest near step 1000 and rises after, so these runs stop at 1500 steps. The TinyStories runs take 328M and 655M tokens (about 0.6 and 1.2 epochs).

Optimizers. Write g_t for the mini-batch gradient at step t and $\eta_t = \eta\,\varphi(t)$ for the learning rate, with $\varphi(t) = (t+1)/w$ for $t < w$ and $\varphi(t) = 0.1 + 0.45\,(1 + \cos(\pi\,(t-w)/(T-w)))$ after, T being the last step.

- **SGD** with heavy-ball momentum $\mu = 0.9$ and no weight decay: $b_{t+1} = \mu b_t + g_t$, $\theta_{t+1} = \theta_t - \eta_t b_{t+1}$.
- **AdamW** with $(\beta_1, \beta_2) = (0.9, 0.95)$ ($(0.9, 0.99)$ for Shakespeare, as in nanoGPT), $\epsilon = 10^{-8}$, and decoupled weight decay 0.1 on every tensor with at least two dimensions (including embeddings and head) and none on LayerNorm gains: $\theta_{t+1} = (1 - \eta_t \lambda_{\text{wd}})\theta_t - \eta_t\,\hat{m}_{t+1}/(\sqrt{\hat{v}_{t+1}} + \epsilon)$, with bias-corrected moments $\hat{m}, \hat{v}$.
- **Muon** (Jordan et al., 2024) on every matrix inside the transformer blocks (query, key, value, attention output, both MLP matrices), with momentum $\beta = 0.95$, Nesterov, and no weight decay: $M_{t+1} = \beta M_t + (1-\beta)G_t$, $W_{t+1} = W_t - \eta_t \sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}\,\text{NS}_5\big((1-\beta)G_t + \beta M_{t+1}\big)$, where NS_5 is five steps of the quintic Newton–Schulz iteration that approximates the orthogonal polar factor. Embeddings, head and gains are updated by an auxiliary AdamW with $(0.9, 0.95)$, $\epsilon = 10^{-8}$, no weight decay, and its own learning rate (Table 1).

Gradient clipping is off, since it would hide edge-of-stability dynamics. Training steps of the language models use bfloat16 autocast; all measurements (Section 4) are in float32.

Learning rates. For each setting and optimizer we ran five learning rates (grids in Table 4) on data order 0, with the schedule and length of the main runs and no measurements, and kept the one with the best final validation loss (on the validation sequences of the probe, Section 4.1). Shakespeare is the exception: since its runs overfit, we kept the best *minimum* validation loss over training. SGD is the exception to the rule: in every setting its best rate on data order 0 diverged or blew up on other data orders (0.2 on Shakespeare, 0.1 on TinyStories-10M, 0.3 on TinyStories-30M), so SGD uses a lower rate: 0.05 on Shakespeare (the smallest of its grid) and the next lower rate on TinyStories, 0.03 and 0.1. On Shakespeare every SGD rate of the grid stops at about the same minimum validation loss (2.44–2.50, the bigram level), so we took the smallest and most stable one. Runs whose training loss became non-finite were stopped and counted as diverged.

Shared initialization and data order. Every run starts from the same initialization (seed 0). The mini-batch at step t depends only on the data order d and on t: for the language models it is B windows of 256 tokens at offsets drawn

uniformly from the training split by a generator seeded with (d, t); for CIFAR-10 it is a slice of a per-epoch permutation seeded with (d, epoch), dropping the last incomplete batch. Two runs with the same d therefore see identical batches whatever their optimizer, and forks and resumed runs see the batches their parent would have seen. Batches used for measurements come from separate random streams and never change the training order. Each optimizer is run with data orders $d \in \{0, \dots, 4\}$: 15 main runs per setting. The runs with $d = 0$ also keep the other optimizers' states in shadow and save fork checkpoints (Section 4.5).

Control for Muon's auxiliary Adam. On Shakespeare, Muon's auxiliary AdamW and the AdamW runs differ in learning rate (10^{-3} against $3 \cdot 10^{-4}$), β_2 (0.95 against 0.99) and weight decay (none against 0.1). A control suite repeats Muon with its auxiliary AdamW matched to AdamW (learning rate $3 \cdot 10^{-4}$, $\beta = (0.9, 0.99)$, weight decay 0.1, which here also applies to the gains), with data orders 0–4 ("matched Muon"). It differs from AdamW only in how the hidden matrices are trained (and in decay on the gains).

4 Methods

Step s always refers to the model θ_s after s updates ($s = 0$ is the initialization). Each run logs predictions at its prediction steps $\mathcal{S}$ and curvature at its measurement steps $\mathcal{M}$; both grids combine a linear and a log-spaced grid (Table 6).

4.1 Function distance and alignment

Distance. Every run stores, at each $s \in \mathcal{S}$, its next-token distributions on a fixed set of N held-out positions, identical across runs: for Shakespeare all 256 positions of 64 validation sequences ($N = 16{,}384$), for TinyStories 16 evenly spaced positions of 128 validation sequences ($N = 2{,}048$), and for CIFAR-10 the 10,000 test images. Write $p_i^X(s)$ for the distribution of run X at step s on position i. The distance between run X at step s and run Y at step s' is the mean Jensen–Shannon divergence in bits,

$$D_{\mathrm{JS}}(X_s, Y_{s'}) = \frac{1}{N} \sum_{i=1}^{N} \mathrm{JS}\big(p_i^X(s),\, p_i^Y(s')\big), \qquad \mathrm{JS}(p, q) = \tfrac{1}{2}\mathrm{KL}(p \,\|\, m) + \tfrac{1}{2}\mathrm{KL}(q \,\|\, m), \quad m = \tfrac{1}{2}(p + q), \tag{1}$$

with base-2 logarithms, so $D_{\mathrm{JS}} \in [0, 1]$. The second distance is the top-1 disagreement $D_{\mathrm{dis}}(X_s, Y_{s'}) = 1 - \frac{1}{N}\sum_i \mathbf{1}\{\arg\max_v p_{i,v}^X(s) = \arg\max_v p_{i,v}^Y(s')\}$.

Alignment by step. $D(X_s, Y_s)$ at the steps $s \in \mathcal{S}$ that both runs log.

Alignment by training loss. Optimizers progress at different speeds, and comparing at equal steps confounds "faster" with "different". Each run therefore also logs, at each $s \in \mathcal{S}$, its mean cross-entropy $\ell_X(s)$ on a fixed training probe, the same for all runs (64 training sequences, all positions, for the language models; 2,000 training images for CIFAR-10). We make it monotone with a running minimum, $\bar{\ell}_X(s) = \min\{\ell_X(s') : s' \in \mathcal{S},\ s' \le s\}$. Each step s of X is matched with the earliest step of Y whose loss is closest,

$$\pi_{XY}(s) = \min\Big(\arg\min_{s' \in \mathcal{S}} \big|\bar{\ell}_Y(s') - \bar{\ell}_X(s)\big|\Big), \qquad \text{kept if } \big|\bar{\ell}_Y(\pi_{XY}(s)) - \bar{\ell}_X(s)\big| \le \tau\, \bar{\ell}_X(s), \tag{2}$$

with $\tau = 0.02$. The kept matches give points $\big(\bar{\ell}_X(s),\, D(X_s, Y_{\pi_{XY}(s)})\big)$, and the *loss-matched distance* $D^{\mathrm{loss}}_{X,Y}(\ell)$ is their piecewise-linear interpolation in ℓ (where several steps share a loss value, the earliest is used), undefined outside the range of kept losses. The loss coordinate is that of the first run, X.

Noise floor. Two runs of the same optimizer from the same initialization with different data orders, $D^{\mathrm{loss}}_{O_d, O_{d'}}$ with $d \neq d'$, differ only through the sampling of mini-batches. A difference between optimizers counts only where it exceeds this floor.

4.2 Locating the split

Fix two optimizers A and B and write A_d, B_d for their runs with data order $d \in \mathcal{D} = \{0, \dots, 4\}$. Runs with the same d share initialization and batches, so they form the *cross* pairs; runs of one optimizer with different d form the *floor* pairs:

$$c_d(\ell) = D^{\mathrm{loss}}_{A_d, B_d}(\ell), \qquad f^O_{dd'}(\ell) = D^{\mathrm{loss}}_{O_d, O_{d'}}(\ell) \quad (O \in \{A, B\},\ d < d').$$

Loss grid. Let $\ell_1 > \ell_2 > \cdots > \ell_J$ be the distinct values of $\bar{\ell}_{A_0}(s)$, $s \in \mathcal{S}$, that lie in the range covered by all $2|\mathcal{D}|$ runs,

$$\Big[\max_X \min_{s\in\mathcal{S}} \bar{\ell}_X(s),\ \min_X \max_{s\in\mathcal{S}} \bar{\ell}_X(s)\Big].$$

Decreasing loss is the order of training.

Statistic. For a multiset S of data orders with distinct elements $U(S)$, let

$$\bar{c}_S(\ell) = \mathrm{mean}_{d\in S}\ c_d(\ell), \qquad F_S(\ell) = \max\Big(\mathrm{mean}_{d<d'\in U(S)}\ f^A_{dd'}(\ell),\ \mathrm{mean}_{d<d'\in U(S)}\ f^B_{dd'}(\ell)\Big), \tag{3}$$

where the cross mean counts repeated orders with their multiplicity, means skip undefined values, and the maximum takes the defined side if only one is. The floor is thus the larger of the two optimizers' average seed-to-seed distances. A level is *above the floor* if the cross distance exceeds the floor by a margin,

$$o_S(j) = \mathbf{1}\big\{\bar{c}_S(\ell_j) > (1+m_\times)\, F_S(\ell_j) + m_+\big\}, \tag{4}$$

with $m_\times = 0.5$ and $m_+ = 0$; a comparison with an undefined value counts as false.

First exceedance. The first grid level from which the cross distance is above the floor at $k = 3$ consecutive levels,

$$t^*(S) = \ell_{j^*}, \qquad j^* = \min\big\{j :\ o_S(i) = 1 \ \text{ for } i = j, \ldots, j+k-1\big\}, \tag{5}$$

undefined if no such j exists. Since the grid is made of the losses of A_0 at its prediction steps, the k levels are consecutive distinct loss values of that run. This was the planned definition of the split, but it is degenerate: at initialization all runs compute the same function, so the floor is close to 0 for the first steps, while two different optimizers move apart from their first update. In every pair of optimizers t^* is one of the first grid levels, reached at steps 1–8 (Table 2). It records that two optimizers differ from the start, not that they split. We therefore also report:

Episodes and persistent split. An *episode* is a maximal run of consecutive levels $\ell_{j_0}, \ldots, \ell_{j_1}$ with $o_S(j) = 1$ that contains at least k levels ($j_1 - j_0 + 1 \geq k$), reported as the loss range $[\ell_{j_0}, \ell_{j_1}]$. The *persistent split* $t^*_{\mathrm{p}}(S)$ is the start ℓ_{j_0} of the last episode if that episode ends at the last level of the grid ($j_1 = J$), and undefined otherwise: the loss from which the cross distance stays above the floor down to the lowest loss that all runs reach. A persistent split is what "together first, apart later" predicts; an episode that ends before ℓ_J is a difference that later disappears. Figures also show the ratio $\bar{c}_{\mathcal{D}}(\ell)/F_{\mathcal{D}}(\ell)$ against the loss, with the threshold $1 + m_\times$.

Estimate and uncertainty. The point estimates are $t^* = t^*(\mathcal{D})$, $t^*_{\mathrm{p}} = t^*_{\mathrm{p}}(\mathcal{D})$ and the episodes of $\mathcal{D}$. For their uncertainty we draw 1000 bootstrap resamples S_b of $|\mathcal{D}|$ data orders with replacement and report, for t^* and t^*_{p} separately, the 2.5 and 97.5 percentiles of the defined values, together with the fraction of resamples in which one is found (a resample with a single distinct order has no floor and gives none). The distance is D_{JS}; D_{dis} and other values of $m_\times$, k and τ are checked in Section 5.2 (300 resamples each).

In steps. t^* is a loss level, which makes it comparable across optimizers that reach it at different steps. For each run X, $t^*_X = \min\{s \in \mathcal{S} : \bar{\ell}_X(s) \leq t^*\}$ is the first step at which X reaches it, and likewise $t^*_{\mathrm{p},X}$ for the persistent split.

4.3 Optimizer against data order, and n-gram references

Decomposition at equal steps. For two optimizers X, Y with data orders $d = 0, \ldots, n-1$ ($n = 5$) and $d^+ = (d+1) \bmod n$, we compare at each shared prediction step s four families of n pairs: $D(X^d_s, Y^d_s)$ (other optimizer, same data: the cross pairs above), $D(X^d_s, Y^{d^+}_s)$ (other optimizer, other data), and the floors $D(X^d_s, X^{d^+}_s)$ and $D(Y^d_s, Y^{d^+}_s)$ (same optimizer, other data), and plot the median and range over d. If the same-data distance is no larger than the floors, the change of optimizer changes the function no more than a change of data order; if it is smaller than the other-data distance, sharing the batches keeps the two optimizers closer.

N-gram references. From the counts $n(b)$ and $n(a,b)$ of tokens and of consecutive token pairs in the training split we build a unigram model $p_1(b) \propto n(b)+1$ and a bigram model smoothed towards it, $p_2(b \mid a) = \big(n(a,b)+\alpha\, p_1(b)\big)/\big(n(a)+\alpha\big)$ with $\alpha = 1$, evaluated at the validation probe positions (the bigram conditioned on the current token). For a run X we plot D_{JS} (1) between $p^X(s)$ and each reference against the training loss $\bar{\ell}_X(s)$. A minimum near zero means the run passes through that n-gram solution; two optimizers that pass through it at the same loss meet there in function space.

4.4 Sharpness and stability thresholds

At each $s \in \mathcal{M}$, before the update of that step, we measure on θ_s (in float32):

Sharpness. $\lambda_{\max}$ is the largest eigenvalue of the Hessian $H = \nabla^2 L_{\mathcal{H}}(\theta_s)$ of the mean cross-entropy on a fixed batch $\mathcal{H}$ of training examples (32 sequences of 256 tokens for the language models; 2,500 images, 5% of the training set, for CIFAR-10), computed by Lanczos with full reorthogonalization (20 iterations from a fixed random start) on Hessian-vector products.

SGD threshold. On a quadratic with curvature λ, heavy-ball momentum as above is stable if and only if $\eta\lambda < 2+2\mu$. We compare $\lambda_{\max}(\theta_s)$ with $(2+2\mu)/\eta_s$, which is $3.8/\eta_s$ for $\mu = 0.9$ and $2/\eta$ for the plain SGD of the break-even runs.

AdamW preconditioned sharpness. Following Cohen et al. (2022), let $P = (1-\beta_1^t)\,\mathrm{diag}(\sqrt{\hat{v}}+\epsilon)$, with $\hat{v} = v/(1-\beta_2^t)$ built from the optimizer's current second-moment state and step count t. The AdamW update is then $-\eta_t P^{-1} m$ with m the uncorrected first moment, so with P frozen and weight decay ignored it is heavy-ball momentum with coefficient β_1 and step $\eta_t(1-\beta_1)$ in the metric P. It is stable if $\lambda_{\max}(P^{-1}H) < (2+2\beta_1)/((1-\beta_1)\eta_t)$, which is $38/\eta_t$ for $\beta_1 = 0.9$. We compute the preconditioned sharpness $\lambda_{\max}(P^{-1/2}HP^{-1/2})$, which has the spectrum of $P^{-1}H$, by the same Lanczos procedure, for every run that trains with AdamW (including forks into AdamW).

Batch sharpness. Following Andreyev and Beneventano (2024, Def. 1),

$$\mathrm{BS}(\theta) = \frac{\sum_{j=1}^{n} g_j^\top H_j\, g_j}{\sum_{j=1}^{n} \|g_j\|^2}, \tag{6}$$

where g_j and H_j are the gradient and Hessian of the loss on mini-batch j, over $n = 8$ random training mini-batches of the training batch size (a fresh random stream at every measured step). For mini-batch SGD without momentum, Andreyev and Beneventano (2024) find that BS stabilizes at $2/\eta$. We measure it for every optimizer and compare it with $(2+2\mu)/\eta_s$ for SGD.

Muon. We have no stability threshold for Muon, and report its $\lambda_{\max}$ and batch sharpness without one. Its update on a hidden matrix is orthogonalized, with all singular values near one whatever the size of the gradient, so the quadratic argument above, where the step scales with the gradient, does not carry over directly. This is a limitation. What we can test is the part of the model that Muon leaves to its auxiliary AdamW (Section 4.4).

One-step loss change. $\Delta L(s) = \ell_X(s) - \ell_X(s+1)$ on the training probe, used in the break-even analysis.

Edge of stability of the Adam-managed block. Muon trains the embeddings, head and gains with an auxiliary AdamW; write a for these coordinates. At the fork checkpoints of the $d = 0$ runs (Table 5), which hold the optimizer states, we compute the top eigenvalue of $P_a^{-1/2} H_{aa} P_a^{-1/2}$, where H_{aa} is the block of the Hessian on $\mathcal{H}$ restricted to a and P_a the preconditioner P defined above, built from the state of the run's own Adam for these coordinates (Muon's auxiliary AdamW, or AdamW itself), by the same Lanczos procedure. We compare it with $(2+2\beta_1)/((1-\beta_1)\eta_s)$, with η and β_1 of that Adam. This treats the block as if the rest of the model were frozen, so it is a check of the auxiliary AdamW on its own, not a threshold for Muon.

What coincides with the split. For a sharpness measure $\mathcal{Q}$ with threshold Θ, the measured steps at the edge of stability are $\mathcal{E}(X) = \{s \in \mathcal{M} : \mathcal{Q}_X(s) \geq (1-\delta)\,\Theta_X(s)\}$ with $\delta = 0.1$. We report the onset $s_{\mathrm{EoS}}(X) = \min \mathcal{E}(X)$, the last step of $\mathcal{E}(X)$, and the fraction $|\mathcal{E}(X)|/|\mathcal{M}|$ of measured steps at the edge of stability, for $\lambda_{\max}$ (SGD) and the preconditioned sharpness (AdamW), and the ratio $\mathcal{Q}_X(s)/\Theta_X(s)$ at the steps t_X^* and $t_{\mathrm{p},X}^*$. Since $\mathcal{M}$ is partly log-spaced, the fraction weights early training more than a fraction of steps would. Batch sharpness is shown in the figures but has no onset computed. We compare these with t_X^* and $t_{\mathrm{p},X}^*$, with the episodes, with the end of warmup w, and with the loss levels across settings.

4.5 Swaps with warm optimizer states

Shadow states. During each $d = 0$ run of optimizer A, the other two optimizers are stepped on the same mini-batch gradients with learning rate 0. The state updates of SGD momentum, AdamW and Muon do not depend on the learning rate, and with learning rate 0 neither the update nor the decoupled weight decay moves the parameters; so their momenta, second moments and step counts evolve as if they were training, while the parameters follow A alone. (Our copy of Muon does not overwrite the gradient in place, so a shadow optimizer stepping after it sees the same gradient.)

Forks. At fork steps s (nine per setting, Table 5), the $d = 0$ run saves θ_s and the states of all three optimizers. The fork $A \to B$ at s loads θ_s and B's shadow state, keeps B's own hyperparameters and learning rate, and continues from s to T on the same schedule φ and the same batches. All six ordered pairs are forked at all nine steps. The control "A continued with the same batches" is the parent run itself. In addition, A is forked into A with data order 1 at the same steps, the control for mode connectivity (Section 4.6). This gives 81 forks per language-model setting. Forks log predictions and measurements like their parents.

Where a fork lands. Let F be the final model of the fork $A \to B$ at s, A the final model of the parent A_0, and B the final model of the pure run B_0 (from initialization, same batches). With D either distance of Section 4.1, the landing coordinate is

$$r = \frac{D(F, A)}{D(F, A) + D(F, B)}, \tag{7}$$

which is 0 at A's function and 1 at B's. It is a ratio of distances, not a projection: a fork far from both ends has r near $1/2$, so we report $D(F, A)$, $D(F, B)$ and $D(A, B)$ with it. For each function metric ϕ of Section 4.8 we also report the shift $(\phi(F) - \phi(A))/(\phi(B) - \phi(A))$, the fraction of the gap between the pure runs that the swap closes. If the function is decided at the split, forks with $s < t^*_A$ should end near $r = 1$ and forks with $s > t^*_A$ near $r = 0$. Since t^* turns out to be degenerate (Section 4.2), the time to compare fork steps with is the end of the early episode.

4.6 Linear mode connectivity

For two final weight vectors θ_1, θ_2 of the same architecture let $L(\alpha)$ be the mean cross-entropy of $(1 - \alpha)\theta_1 + \alpha\theta_2$ (every floating-point entry of the state interpolated) on the 64 sequences of the training probe. The barrier is

$$\mathrm{Bar}(\theta_1, \theta_2) = \max_{\alpha \in \{0, 0.1, \ldots, 1\}} \big[L(\alpha) - (1 - \alpha)L(0) - \alpha L(1)\big], \tag{8}$$

a loss version of the instability of Frankle et al. (2020). We compute it between the parent A_0 and each of its forks, as a function of the fork step: the forks $A \to A$ with data order 1 reproduce the instability analysis of Frankle et al. (2020) for each optimizer, and the forks $A \to B$ ask whether a change of optimizer at s leaves the solution in the same basin.

4.7 Anatomy of the top eigenvector

Partition the parameters into blocks: *emb* (token and position embeddings); for each layer i, qk_i (W_Q, W_K), ov_i (W_V and the attention output W_O) and mlp_i (both MLP matrices); *norms* (all LayerNorm gains); and *head*. For the CNN each layer is a block. Let v be the unit top eigenvector of H returned by Lanczos, and v_b the vector equal to v on the coordinates of block b and zero elsewhere. Since the blocks partition the coordinates, $v = \sum_b v_b$ and $\sum_b \|v_b\|^2 = 1$, and expanding the quadratic form gives

$$\lambda_{\max} = v^\top H v = \sum_b s_b + \chi, \qquad s_b = v_b^\top H_{bb} v_b, \qquad \chi = \sum_{b \neq c} v_b^\top H_{bc} v_c, \tag{9}$$

where H_{bc} is the block of H with rows in b and columns in c (and v_b is read as its restriction to block b where needed). We record, for each block, its share of the direction $\|v_b\|^2$ and its self-curvature s_b, and the cross-block remainder $\chi = v^\top H v - \sum_b s_b$, with $v^\top H v$ from one more Hessian-vector product (it equals the Lanczos estimate of $\lambda_{\max}$ up to the Lanczos error). Both s_b and χ can be negative, and $s_b \leq \lambda_{\max}(H_{bb})\|v_b\|^2$: a block carries a large part of the sharpness only if the direction lies in it and its own diagonal block is curved along it, while a large χ means the sharp direction comes from interactions between blocks. The split of the project plan, $v_b^\top (Hv)_b$, equals $\lambda_{\max}\|v_b\|^2$ for an exact eigenvector and so repeats the direction share; this is why we use (9) instead. For AdamW runs we repeat the decomposition for $P^{-1/2} H P^{-1/2}$ and its top eigenvector, whose diagonal blocks are $P_b^{-1/2} H_{bb} P_b^{-1/2}$. Figures sum blocks by type (all *qk*, all *ov*, all *mlp*) or by layer.

4.8 Function metrics

Offline, on saved checkpoints, we compute three groups of metrics at fixed validation points (definitions in Section B.1). *Input derivatives* of the loss as a function of the input (the token embeddings for the language models), from first to third order: gradient norm, Jacobian norm of the softmax output (Novak et al., 2018), Frobenius norm and top eigenvalue of the input Hessian (Moosavi-Dezfooli et al., 2019), level-set curvature (Srinivas et al., 2022), and three third-order quantities. *Weight norms*: those linear theory predicts each optimizer keeps small (ℓ_2, ℓ_∞, spectral), and

norm-based measures compared by Jiang et al. (2020). *Accuracy and robustness*: Gaussian input noise for the CNN (Ford et al., 2019) and one-token substitution sensitivity (Hahn et al., 2021) for the language models.

4.9 Break-even replication

We follow the protocol of Jastrzebski et al. (2020): plain SGD (no momentum) at a constant learning rate, without warmup. Besides $\lambda_{\max}$ (their λ_H^1) and ΔL, at each measured step we draw $L = 25$ mini-batch gradients $g_1, \ldots, g_L$ and estimate the gradient covariance $\hat{K} = \frac{1}{L}\sum_i (g_i - \bar{g})(g_i - \bar{g})^\top$ through the eigendecomposition of its $L \times L$ Gram matrix. We record its top eigenvalue λ_K^1, its smallest non-zero eigenvalue λ_K^* ($\hat{K}$ has rank at most $L - 1$), $\operatorname{Tr} \hat{K}$, and $\|g\|/\|g_5\|$, the mean over i of $\|g_i\|/\|\Pi_5 g_i\|$ with Π_5 the projection on the top five eigenvectors of $\hat{K}$. Prediction trajectories are embedded in two dimensions (PCA, then UMAP; Section B).

Runs. CIFAR-10: learning rates 0.001, 0.01 and 0.1 at batch size 128 (39,000 steps, 100 epochs), and batch sizes 30 and 300 at learning rate 0.05 (also 100 epochs), with $\hat{K}$ always from batch-128 gradients; two data orders each. Shakespeare and TinyStories-10M: learning rates 0.03, 0.1 and 0.3 at the training batch size, with the setting's model and length; three data orders each. A rate of 1.0 diverged within 15 (Shakespeare) and 20 (TinyStories-10M) steps and was dropped. The split analysis of Section 4.2 applied to two learning rates instead of two optimizers measures the separation of trajectories in prediction space.

5 Results

5.1 Learning-rate sweeps

Figure 8 (Section D) shows the sweeps. In all three settings Muon reaches the lowest validation loss, then AdamW, then SGD: best final validation loss 1.305, 1.339 and 1.914 on TinyStories-10M and 1.181, 1.201 and 1.340 on TinyStories-30M (SGD at its best rate on data order 0; 1.957 and 1.370 at the rates we use), and best minimum validation loss 1.54, 1.56 and 2.44 on Shakespeare (2.50 for SGD at 0.05, about the loss at which all three optimizers pass the bigram model, Section 5.3). SGD at 1 and 3 diverged on both TinyStories models. Without dropout the Shakespeare runs overfit, Muon fastest: its validation loss is lowest near step 500 and ends at 2.84–2.88 in the main runs, against 1.54–1.55 for AdamW (Fig. 9).

5.2 Divergence curves and the split

Table 2 summarizes the split analysis for all pairs and Fig. 1 shows three of them. The same pattern holds in every language-model pair.

The optimizers differ from the first steps. The first exceedance t^* is the first or second level of the loss grid, reached at steps 1–8. At that point the floor is still near zero, while two optimizers have already taken different first steps.

An early episode, then back to the floor. Each pair has one long early episode above the floor. It ends at loss 4.59–6.82 on TinyStories and 2.77–3.16 on Shakespeare, and the ratio peaks in it at 32–204. The ratio is large because the floor is near zero; in absolute terms the cross distance in the episode is at most 0.034–0.045 bits for AdamW–Muon and 0.04–0.18 bits for pairs with SGD. After the episode the ratio returns to near 1 and stays there. For AdamW–Muon it lies in 0.95–1.38 (median 1.02) below loss 4.85 on TinyStories-10M, in 0.87–1.34 (median 1.01) below loss 4.59 on TinyStories-30M, and in 1.02–1.46 (median 1.11) below loss 2.12 on Shakespeare, after a second, short episode (2.30–2.12). At the lowest common loss all TinyStories pairs are at 0.99–1.03 times the floor. The largest absolute AdamW–Muon distances, 0.14–0.15 bits near loss 2.3 on TinyStories, occur where the floor is as large (middle panels).

One persistent split. Only Muon–SGD on Shakespeare has a persistent split: a second episode from loss 2.82 to the end of the common grid at 2.61, found in 96.5% of resamples (95% interval 2.75–4.25). Muon reaches it at steps 14–17 and SGD at steps 43–550, depending on the data order. This pair tells us the least about how optimizers shape the function: the comparison stops at 2.61 because the slowest SGD run gets no further, SGD at 0.05 is barely stable there (loss spikes, Fig. 9; erratic floor, Fig. 1 bottom left), and four of its five runs end at training loss 2.32–2.58, about the bigram level (the fifth at 1.66). AdamW–SGD on Shakespeare is borderline: after its episode the ratio stays at 1.33–1.73, which is a persistent split in 34% of resamples but not in the full sample. On TinyStories SGD's own data-order noise is large (on TinyStories-30M one run ends at validation loss 1.61 against 1.37–1.39 for the others; on

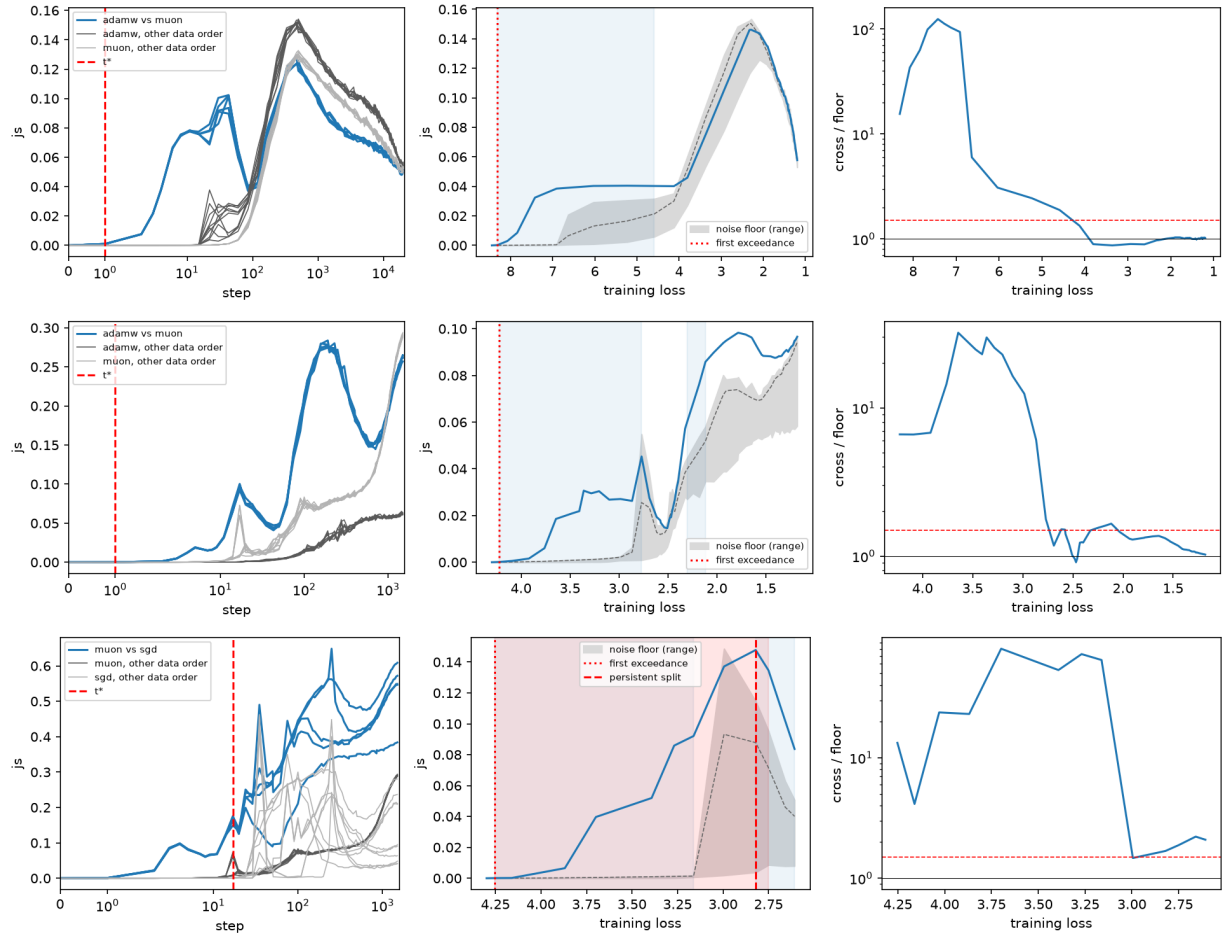


Figure 1: Function distance D_{JS} (bits) between optimizers. Rows: TinyStories-30M AdamW–Muon, Shakespeare AdamW–Muon, Shakespeare Muon–SGD (the only persistent split). Left: by step, for the 5 cross pairs (blue, same data order) and the 10 floor pairs of each optimizer (greys); the red line is the median step at which the first runs reach the persistent split, or the first exceedance t^* if there is none. Middle: loss-matched distance (blue) against training loss (axis reversed), with the range over floor pairs (grey band), the floor $F_{\mathcal{D}}$ (dashed), the episodes above the floor (shaded), t^* (dotted) and the persistent split (dashed, with its 95% interval shaded red). Right: cross distance over the floor, with the threshold $1 + m_{\times} = 1.5$ (red). The other pairs are in Figs. 10 and 11.

TinyStories-10M the runs end at 1.77–1.96), so "no split" against SGD means no larger than SGD's own data-order noise.

By step the picture differs. At equal steps (left panels) the optimizers look different for much longer, because they progress at different speeds: on TinyStories-30M Muon's loss stays flat for about the first ten steps and then falls faster than AdamW's, and on Shakespeare Muon memorizes the training set while AdamW does not (Fig. 9). Section 5.3 separates the two effects.

Robustness. We repeated the estimate for every pair and setting with D_{dis} instead of D_{JS}, and with D_{JS} at $m_{\times} \in \{0.25, 1\}$, $k \in \{2, 5\}$ and $\tau \in \{0.01, 0.05\}$, one change at a time (seven variants). On both TinyStories models no variant finds a persistent split for any pair, nor does any variant for AdamW and Muon on Shakespeare. The one persistent split, Muon against SGD on Shakespeare, is fragile: it is found with $k = 2$ (99% of resamples), $m_{\times} = 0.25$ and both tolerances, but not at the point estimate with D_{dis}, with $m_{\times} = 1$ (26% of resamples) or with $k = 5$ (42%). We read it as SGD ending slightly off the others where it stalls, not as a clear split.

5.3 Where the optimizers' routes meet

Optimizer against data order. On TinyStories-30M (Fig. 2, top left), from step 332 on, the AdamW–Muon distance on the same data order is at or below both optimizers' floors. At step 20,000 it is 0.051 bits, against floors of

Table 2: Split analysis with D_{JS}. Grid: range of the common loss grid. t^*: first exceedance (5), found in 99.8% of resamples for every language-model pair (all resamples with at least two distinct data orders) and 52% for CIFAR-10 (two data orders), with a 95% interval equal to the point; steps: the steps at which A_0 and B_0 reach it. Episodes: loss ranges above the floor. Persistent: the persistent split t^*_{p} and the fraction of resamples in which one is found. Max and end: the ratio $\bar{c}_{\mathcal{D}}/F_{\mathcal{D}}$ at its maximum and at the last grid level. "Matched": Muon with its auxiliary AdamW matched to AdamW (Section 3). Break-even rows compare plain SGD at two learning rates.

Setting	Pair A–B	Grid	t^*	steps	Episodes	Persistent	Max	End
Shakespeare	AdamW–Muon	4.30–1.18	4.23	1 / 2	4.23–2.77, 2.30–2.12	none (0%)	32	1.02
	AdamW–SGD	4.30–2.62	4.23	1 / 1	4.23–3.10	none (34%)	98	1.69
	Muon–SGD	4.30–2.61	4.25	1 / 1	4.25–3.16, 2.82–2.61	**2.82** (96.5%)	80	2.09
	Matched–AdamW	4.30–1.19	4.26	1 / 1	4.26–2.85, 2.70–1.68	none (0%)	29	1.07
	Matched–Muon	4.30–0.23	none (4%)	–	none	none (0%)	1.28	0.98
TinyStories-10M	AdamW–Muon	8.39–1.35	8.26	1 / 7	8.26–4.85	none (0%)	111	0.99
	AdamW–SGD	8.39–2.04	8.26	1 / 3	8.26–6.32	none (0%)	162	1.03
	Muon–SGD	8.39–2.02	8.38	1 / 1	8.38–5.81	none (0%)	204	1.00
TinyStories-30M	AdamW–Muon	8.45–1.20	8.31	1 / 8	8.31–4.59	none (0%)	124	1.02
	AdamW–SGD	8.45–1.65	8.31	1 / 2	8.31–6.03	none (0%)	121	1.01
	Muon–SGD	8.45–1.65	8.44	1 / 1	8.44–6.82	none (0%)	200	1.00
CIFAR-10	η 0.001–0.01	2.30–1.09	2.27	4290 / 510	4 short, 2.27–1.13	none (0%)	19	1.47
	η 0.01–0.1	2.30–0.0004	0.50	11310 / 2730	0.50–0.41	none (0%)	1.75	1.24
Shakespeare	η 0.1–0.3	4.30–3.09	none (0%)	–	none	none (0%)	1.17	0.72

0.055 (AdamW) and 0.051 (Muon) and 0.056 for AdamW and Muon on different data orders: at the end of training, replacing AdamW by Muon changes the function no more than reshuffling the data. Sharing the data order does keep the two optimizers closer: the same-data distance is 20–32% below the other-data distance between steps 119 and 10,000, and 8% below at the end. Against SGD, the same-data distances at the end are 0.081 (AdamW) and 0.082 bits (Muon), above the median floors (0.051–0.063), and within 5% of the other-data distances; at equal steps this includes SGD being slower (final validation loss 1.37–1.39, one run 1.61, against 1.18–1.20 for AdamW and Muon). On Shakespeare (Fig. 12) the optimizer dominates at equal steps: at step 1500 the AdamW–Muon distance is 0.26 bits, four times AdamW's floor (0.06) but below Muon's own floor (0.29), which grows as Muon memorizes; same and other data orders give the same distance (0.263 and 0.264).

N-gram stages. On Shakespeare (Fig. 2, bottom) the three optimizers pass close to the bigram model at nearly the same training loss: the minimum distance to it is 0.036 (AdamW), 0.041 (Muon) and 0.041 bits (SGD), at loss 2.46, 2.47 and 2.48. At loss 2.47 the loss-matched AdamW–Muon ratio is 0.91. Before that the routes differ: SGD passes through the unigram model (0.001 bits at loss 3.09) and AdamW near it (0.054 at 3.20), while Muon stays further away (0.111 at 2.99). On TinyStories (Fig. 12) the optimizers come closest to the bigram model at loss 3.7–4.0, less closely than on Shakespeare (0.14–0.16 bits for AdamW and Muon, 0.20 for SGD); the AdamW–Muon ratio stays below 1.2 from loss 3.69 (TinyStories-10M) and 3.80 (TinyStories-30M) on. The early episodes therefore cover the stage before the bigram statistics are learned, where the optimizers take different routes, and end around the stage at which they all approximate the same n-gram solution.

5.4 What coincides with the changes in distance

SGD reaches its threshold $(2+2\mu)/\eta_s$ only around the end of warmup (Fig. 3; Table 8 in Section D): in the $d=0$ runs $\lambda_{\max}$ is within 10% of it, or above, only between steps 21 and 120, 62 and 1500, and 60 and 658, for warmups of 100, 200 and 400 steps, and at 2–16% of the measured steps over all data orders. After that, $\lambda_{\max}$ stays below a threshold that rises as the learning rate decays. On Shakespeare the SGD runs have large transient spikes (maximum ratio 7.9–956 over the data orders), matching their loss spikes. Batch sharpness stays below $\lambda_{\max}$, and so below the threshold, apart from early transients (Fig. 13).

AdamW. The preconditioned sharpness is above its threshold at the first measured steps (ratio 3 to 21 at step 1), when the second-moment estimate rests on one or a few gradients, so $s_{\mathrm{EoS}}=1$ in every run. After a dip it returns to near the threshold and stays there for long stretches, until the cosine decay raises the threshold away from it: the ratio is at least 0.9 at measured steps up to 425–600 of 1500 on Shakespeare, 4500–9500 of 10,000 on TinyStories-10M and 9000–19,000 of 20,000 on TinyStories-30M (11–16%, 50–65% and 40–69% of the measured steps). This is the adaptive edge of stability of Cohen et al. (2022).

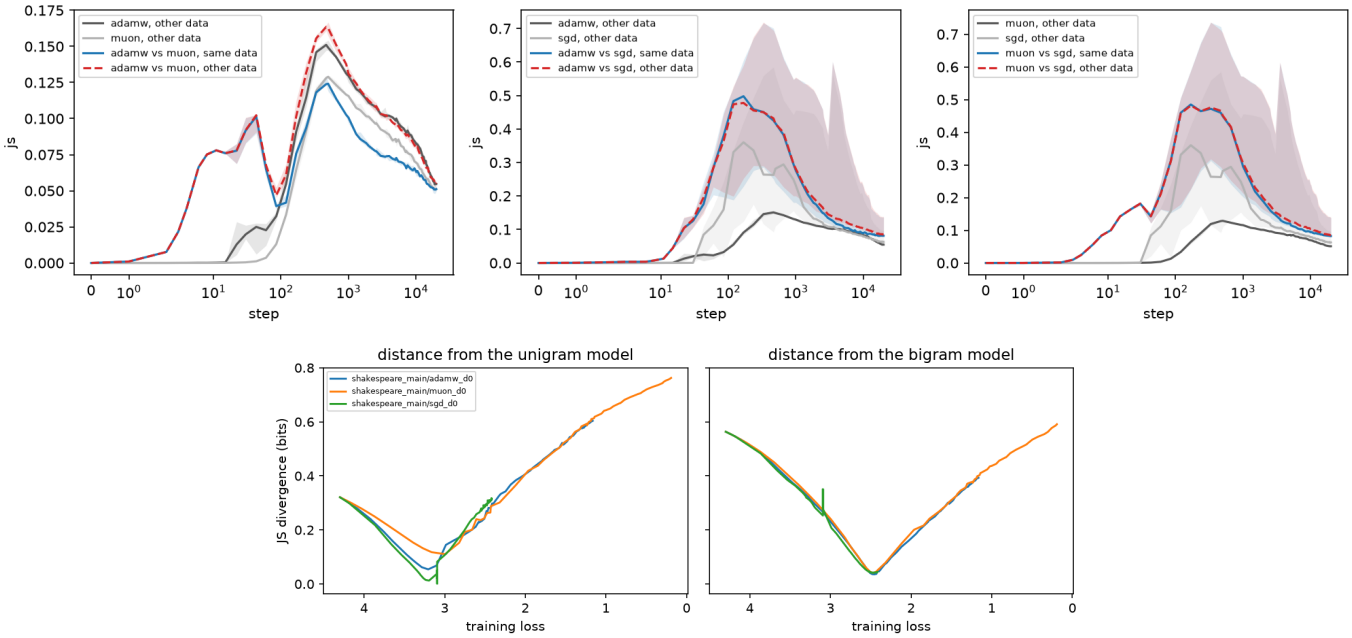


Figure 2: Top: optimizer against data order at equal steps on TinyStories-30M (Section 4.3): other optimizer on the same data order (blue) and on another data order (red, dashed), and the two floors (greys); median over the 5 data orders, range shaded. Bottom: D_{JS} of the $d = 0$ runs on Shakespeare from the unigram (left) and bigram (right) models of the training data, against training loss. Shakespeare decomposition and TinyStories n-gram distances in Fig. 12.

Muon. Without a threshold we only note that its $\lambda_{\max}$ fluctuates widely on Shakespeare (between about 10 and 400) and is smooth on TinyStories (Fig. 13).

Coincidences. No edge-of-stability event explains the changes in distance. The first exceedance, at steps 1–8, comes before the end of warmup and before SGD's onset; it coincides only with AdamW's onset at step 1, which reflects the preconditioner of the first steps rather than an edge of stability. The early episodes are short in steps: the $d = 0$ runs reach the loss at which they end at steps 9–85 (221 for SGD in the TinyStories-10M Muon–SGD pair), within warmup in every pair but that one. For the pairs with SGD this is at SGD steps 20–45 (and 221), in the same range as SGD's onset in the $d = 0$ runs (steps 21, 62 and 60). But the AdamW–Muon pairs return to the floor in the same way without SGD, and AdamW, which reaches that loss at steps 29, 45 and 60, stays near its threshold long after the distance has returned to the floor. At the one persistent split (Muon–SGD, Shakespeare), each SGD run is below its threshold when it reaches loss 2.82 (ratio 0.18–0.80), and Muon's auxiliary AdamW is far below its own (ratio 0.02 at step 25, the first fork checkpoint after Muon reaches 2.82). As a loss level the end of the early episode is similar across the two TinyStories sizes (AdamW–Muon: 4.85 and 4.59) and close to the bigram stage (Section 5.3), which suggests describing it as a stage of learning rather than a step.

The block left to Muon's auxiliary AdamW. On Shakespeare (Fig. 3, bottom left), the embeddings, head and gains of Muon exceed their threshold at the fork checkpoints at steps 300 and 450 (ratio 2.35 and 2.53) and are below it at the other seven (at most 0.57), while in the AdamW run the same block stays at most 0.10 times its threshold. These steps come just before Muon's validation loss starts to rise, and in this period Muon's top Hessian eigenvector lies almost entirely in the embeddings (Section 5.6). On TinyStories-30M it is the other way round: AdamW's block exceeds its threshold at steps 400 and 800 (2.06 and 1.13) and Muon's stays at most 0.53. In both settings the block that overshoots is trained by the Adam with the larger learning rate on these parameters (Shakespeare: Muon's 10^{-3} against AdamW's $3 \cdot 10^{-4}$; TinyStories-30M: AdamW's $2 \cdot 10^{-3}$ against Muon's $1.5 \cdot 10^{-3}$), which points to the learning rate rather than to Muon.

Control. Matching Muon's auxiliary AdamW to AdamW (Section 3) does not remove the memorization: matched Muon still reaches training loss 0.22 with validation loss 2.5, against 1.55 for AdamW, so the memorization comes from Muon's updates of the hidden matrices. [TODO: edge-of-stability ratio of the matched run's auxiliary AdamW (fork checkpoints of `muon_matched_states`) not computed yet.] In function space matched Muon is indistinguishable from Muon (first exceedance in 4% of resamples, ratio at most 1.28; Table 2), and against AdamW it behaves as Muon does: two episodes (4.26–2.85 and 2.70–1.68), no persistent split, and a ratio of 1.07 at the lowest common loss (Fig. 10).

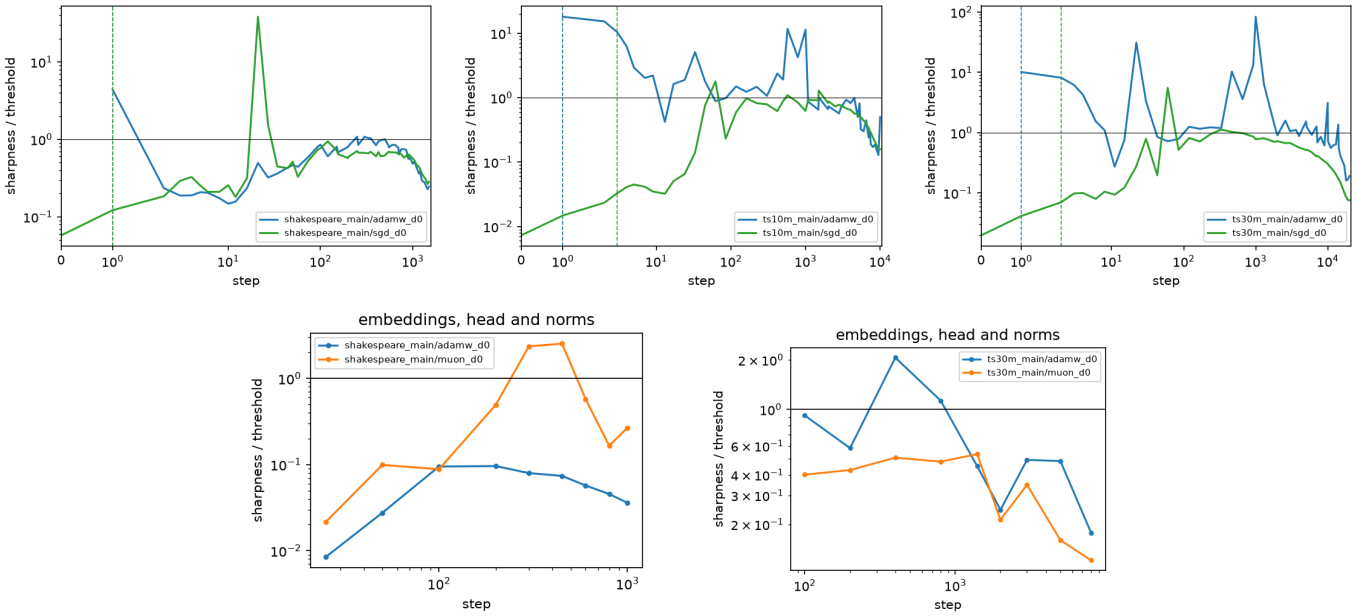


Figure 3: Top: sharpness over its stability threshold for the $d = 0$ runs on Shakespeare, TinyStories-10M and TinyStories-30M: $\lambda_{\max}\,\eta_s/(2+2\mu)$ for SGD and the preconditioned sharpness times $(1-\beta_1)\eta_s/(2+2\beta_1)$ for AdamW; 1 is the edge of stability, dashed lines the first exceedance t^*_X of the AdamW–SGD pair. Bottom: the same ratio for the embeddings, head and gains alone (Section 4.4), at the fork checkpoints of AdamW and Muon, on Shakespeare (left) and TinyStories-30M (right). Raw sharpness, batch sharpness and thresholds in Fig. 13.

[TODO: Swaps: landing coordinate r (7) with D_{JS} against fork step for the six ordered pairs, with the end of the early episode of each pair marked (one panel per setting, or TinyStories-30M here and the rest in the appendix).]

[TODO: Mode connectivity: barrier (8) between A_0 and each fork against fork step; same-optimizer forks with data order 1 (instability control) and cross-optimizer forks.]

Figure 4: Left: where swaps land; 0 is the function of the run that kept its optimizer, 1 that of the run trained with the new optimizer from initialization. Right: loss barrier between the parent and its forks. [TODO: caption details once the swap runs are analysed.]

5.5 Swaps and mode connectivity

[TODO: Describe Fig. 4 (left): r for forks before and after the end of the early episode, for each ordered pair; asymmetries between $A \to B$ and $B \to A$; distances $D(F, A)$, $D(F, B)$ against $D(A, B)$; the function-metric shifts. Since the final AdamW and Muon functions on TinyStories-30M are no further apart than the data-order floor, r for that pair must be read against $D(A, B)$.]

[TODO: Describe Fig. 4 (right): the step after which same-optimizer forks become linearly connected (the Frankle et al. (2020) stability point) for each optimizer; whether cross-optimizer forks stay in the same basin after that point; relation to the early episode.]

5.6 Anatomy of the sharp direction

On TinyStories-30M (Fig. 5) the sharp direction differs by optimizer. Muon's lies mostly in the value–output matrices (about 60–70% of the direction) throughout training, with most of the rest in the MLP matrices. AdamW's lies in the head until about step 4000 and then largely in the gains (30–50%), with the value–output and MLP matrices. SGD's is spread over the MLP, value–output and head blocks. In all three the cross-block term χ is the largest part of the eigenvalue: about 75–85% for Muon and SGD throughout, and for AdamW about 25–40% while the head dominates and 60–70% after. The sharp direction is therefore not the top eigenvector of a single block but comes from curvature between blocks. On TinyStories-10M (Fig. 14) Muon's direction is again in the value–output matrices (60–70%), while AdamW's stays mostly in the head (60–75%). On Shakespeare Muon's direction lies almost entirely (85–97%) in the

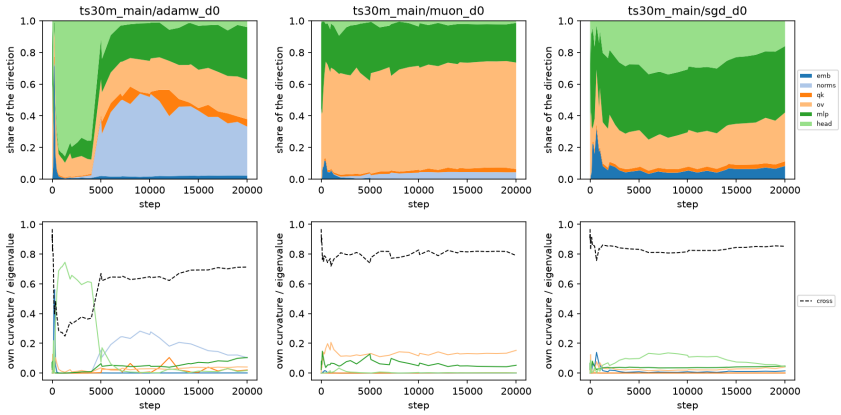


Figure 5: Anatomy of the top Hessian eigenvector of the $d = 0$ runs on TinyStories-30M (9). Top: share of the direction $\|v_b\|^2$ by block type. Bottom: each block type's own curvature s_b and the cross-block remainder χ (dashed), as fractions of the eigenvalue. Shakespeare and TinyStories-10M in Fig. 14.

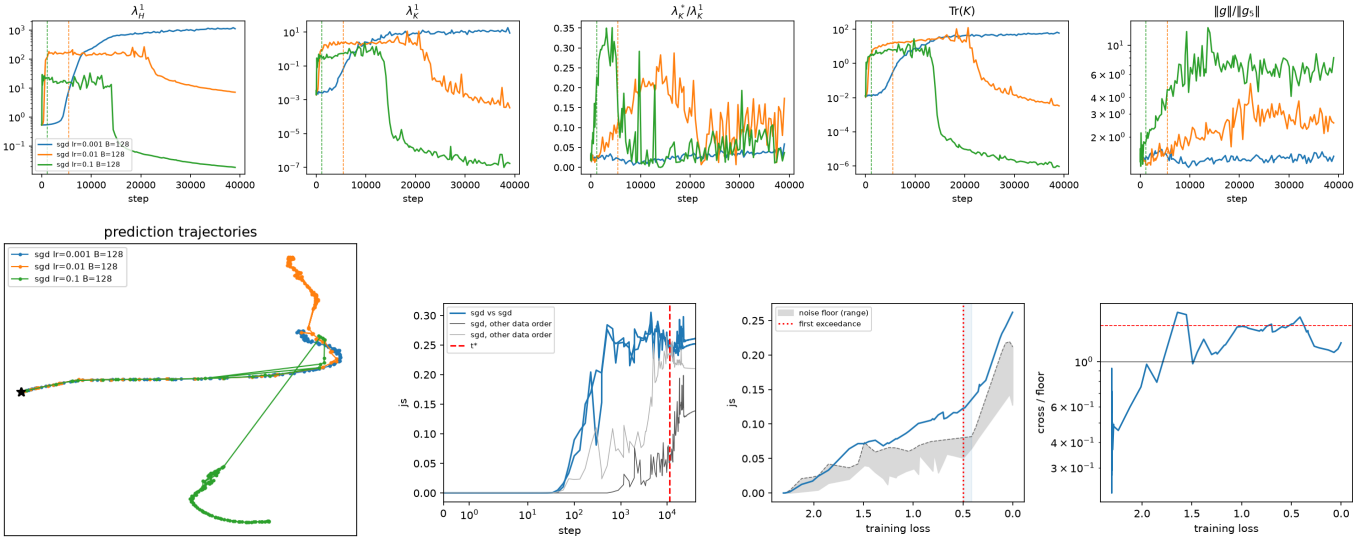


Figure 6: The break-even analysis of Jastrzebski et al. (2020) on CIFAR-10 with plain SGD at $\eta \in \{0.001, 0.01, 0.1\}$, batch 128. Top: λ_H^1, λ_K^1, λ_K^*/λ_K^1, Tr $\hat{K}$ and $\|g\|/\|g_5\|$ against step ($d = 0$). Bottom left: prediction trajectories (PCA then UMAP), starting at the star. Bottom right: the divergence figure of Fig. 1 for $\eta = 0.01$ against $\eta = 0.1$, with two data orders. Shakespeare in Fig. 15.

embeddings from step 200 on, with the embeddings' own curvature making up most of the eigenvalue: the sharp direction is in the block that Muon leaves to its auxiliary AdamW, including the period in which that block is over its threshold (Section 5.4). AdamW's direction on Shakespeare is spread over the MLP, value–output and embedding blocks, with about 70% of the eigenvalue from χ. The changes of dominant block (AdamW on TinyStories-30M, head to gains around step 4000–5000) happen long after the loss-matched distance between optimizers has returned to the floor. [TODO: describe the decomposition of the AdamW preconditioned operator (`anatomy_precond.png`).]

5.7 Break-even replication

CIFAR-10. The effects reported by Jastrzebski et al. (2020) replicate (Fig. 6). The larger the learning rate, the lower the sharpness in the early phase: λ_H^1 levels off at about 1000, 200 and 20 for $\eta = 0.001$, 0.01 and 0.1, that is at or below $2/\eta$, and the maximum of λ_K^1 decreases with the learning rate in the same way. The larger rates also give a better-conditioned gradient covariance (higher λ_K^*/λ_K^1) and a larger $\|g\|/\|g_5\|$. The relation between λ_H^1 and λ_K^1 and ΔL is in Fig. 15. In prediction space the three runs share a path from the initialization; $\eta = 0.1$ leaves it first, then $\eta = 0.01$. At equal steps the distance between $\eta = 0.01$ and 0.1 is far above the data-order floor (bottom right, left panel). At equal training loss, however, it is 1.0–1.75 times the floor at every loss below 1.8 (median 1.24); it exceeds

[TODO: Function metrics against step for the $d = 0$ runs (e.g. top input-Hessian eigenvalue, distance from initialization, sensitivity), with t^* marked.]

Figure 7: Function metrics over training. [TODO: caption details.]

1.5 times the floor at three consecutive levels only once (loss 0.50–0.41) and there is no persistent split. For $\eta = 0.001$ against 0.01 the ratio is 19 near the initialization, where the floor is near zero, and 0.57–2.08 below loss 1.5 (median 1.04), with four short episodes and no persistent split. As for the optimizers, most of the separation in time is one run being faster.

Language models. On Shakespeare, plain SGD at 0.1 and 0.3 without warmup blows up in the first ten or so steps, where the distance between any two runs, floor pairs included, approaches 1 bit; the runs recover but reach only training loss 3.09 (the lowest common loss). At equal loss the distance between the two rates is at most 1.17 times the floor and no split is found (Table 2, Fig. 15). λ_K^*/λ_K^1 rises over training at both rates, and λ_H^1 is lower at 0.3 apart from spikes. [TODO: Shakespeare at $\eta = 0.03$; TinyStories-10M break-even runs still running; CIFAR-10 batch-size runs (30 and 300) not analysed yet.]

5.8 Function metrics

[TODO: Describe Fig. 7 and a table of final values per optimizer (mean over data orders): which metrics separate the optimizers, whether the norms each optimizer is predicted to keep small (ℓ_2, ℓ_∞, spectral) are in fact smallest for it, and the shifts of these metrics under swaps.]

6 Discussion and limitations

Apart first, together later. The hypothesis of the title, that optimizers first follow one path and separate later, is not what we observe. At equal training loss the optimizers are most different at the start, while the data-order floor is still near zero, and the difference disappears around the stage at which all of them approximate the bigram statistics of the data. From there on, on TinyStories, two optimizers on the same data order are as close as one optimizer on two data orders, down to the lowest loss we reach. One reading is that the early phase is governed by the order in which simple statistics are picked up, which depends on the optimizer (on Shakespeare SGD and AdamW pass through the unigram model and Muon does not), while afterwards the function is set mostly by the data and the optimizer matters mainly through speed. This rests on comparing at equal loss: at equal steps all pairs look different for most of training, and on Shakespeare Muon's faster memorization dominates the comparison.

What the floor means. Being within the floor does not mean computing the same function: at the end of the TinyStories runs the AdamW–Muon floor is 0.057–0.067 bits. The finding is that changing the optimizer perturbs the learned function no more than changing the order of the data, not that it does not perturb it. The asymptotic implicit-bias results (Gunasekar et al., 2018; Zhang et al., 2024a; Fan et al., 2025) describe the limit of long training of linear models on separable data. Our TinyStories models see at most 1.2 epochs, and by then the loss-matched difference between optimizers is within the data-order noise. The pending swaps and mode-connectivity tests (Section 5.5) will say whether the optimizer nonetheless decides the endpoint early, as forks into another optimizer could still end at their parent's function.

Break-even point. On CIFAR-10 we reproduce the sharpness and gradient-noise effects of Jastrzebski et al. (2020), but the separation of trajectories between learning rates is mostly a difference in speed: at equal loss, 0.01 and 0.1 stay within 1.75 times the floor. Comparing at equal steps overstates how differently two training procedures shape the function.

Edge of stability. AdamW is at its adaptive edge of stability through much of training whether or not the optimizers differ, and SGD meets its threshold only around the end of warmup; neither explains the start or the end of the early episode, which ends within warmup and in the same way for AdamW and Muon. The one edge-of-stability event that lines up with a change in function, Muon's auxiliary AdamW overshooting its threshold just before Muon memorizes on Shakespeare, is not what drives the memorization, which persists in the matched control. The anatomy adds that the

sharp direction is optimizer-specific and mostly made of curvature between blocks, but its changes do not line up with changes in function distance either.

Limitations.

- All runs share one initialization; the noise floor varies only the data order. Whether the results depend on the initialization is untested.
- Learning rates come from five-point sweeps on one data order, and the Shakespeare AdamW rate is the smallest of its grid. SGD trains poorly at the stable rates on Shakespeare, so comparisons with SGD there stop at loss 2.61, and its large data-order noise on TinyStories makes "no split" against SGD a weak statement.
- The first exceedance is degenerate (Section 4.2); episodes and the persistent split depend on the margin $m_\times$, the run length k and the matching tolerance τ; the conclusions hold over the variants we tried (Section 5.2), except the Shakespeare Muon–SGD split.
- The loss grid ends at the lowest loss that all runs of a pair reach, so a split below it would be missed.
- The thresholds are those of linear stability on a quadratic with a frozen preconditioner; we have no threshold for Muon, and the AdamW ratio at the first steps reflects a preconditioner built from few gradients.
- Sharpness is measured on a fixed batch of 32 sequences with 20 Lanczos iterations, in float32, while training uses bfloat16 autocast.
- Loss matching and the mode-connectivity barrier use a training probe of 64 sequences.
- The landing coordinate is a ratio of distances, not a projection onto the segment between the pure runs.
- The models are small (up to 30M parameters) and trained for at most 655M tokens.

7 Conclusion

From the same initialization and data order, SGD with momentum, AdamW and Muon do not first share a path and then split. At equal training loss they differ most at the start, before data-order noise has built up, and from about the bigram stage on their predictions are as close as those of one optimizer trained on two data orders. This holds in all three language-model settings and for every pair except Muon–SGD on Shakespeare, where SGD barely trains. At the end of TinyStories-30M, replacing AdamW by Muon changes the function no more than reshuffling the data. Neither the edge of stability nor the end of warmup explains these transitions. Whether the optimizer nonetheless decides the endpoint early, that is, whether a run switched from one optimizer to another ends at the function of the first or of the second, is the question the swap and mode-connectivity experiments, still running, will answer.

References

Andreyev, A., A. Ananthkumar, M. Walden, et al. (2026). *Momentum Further Constrains Sharpness at the Edge of Stochastic Stability*. Preprint.

Andreyev, A. and P. Beneventano (2024). *Edge of Stochastic Stability: Revisiting the Edge of Stability for SGD*. Preprint.

Bartlett, P. L., D. J. Foster, and M. Telgarsky (2017). "Spectrally-normalized Margin Bounds for Neural Networks". In: *NeurIPS*.

Cattaneo, M. D., J. M. Klusowski, and B. Shigida (2024). "On the Implicit Bias of Adam". In: *ICML*.

Cohen, J. M., A. Damian, A. Talwalkar, et al. (2025). "Understanding Optimization in Deep Learning with Central Flows". In: *ICLR*.

Cohen, J. M., B. Ghorbani, S. Krishnan, et al. (2022). *Adaptive Gradient Methods at the Edge of Stability*. Preprint.

Cohen, J. M., S. Kaur, Y. Li, et al. (2021). "Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability". In: *ICLR*.

Cortesi, F. V., G. Iannone, G. Crippa, et al. (2026). *Same Error, Different Function: The Optimizer as an Implicit Prior in Financial Time Series*. Preprint.

Damian, A., E. Nichani, and J. D. Lee (2023). "Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability". In: *ICLR*.

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

Fan, C., M. Schmidt, and C. Thrampoulidis (2025). "Implicit Bias of Spectral Descent and Muon on Multiclass Separable Data". In: *NeurIPS*.

Ford, N., J. Gilmer, N. Carlini, and E. D. Cubuk (2019). "Adversarial Examples Are a Natural Consequence of Test Error in Noise". In: *ICML*.

Frankle, J., G. K. Dziugaite, D. M. Roy, and M. Carbin (2020). "Linear Mode Connectivity and the Lottery Ticket Hypothesis". In: *ICML*.

Gunasekar, S., J. Lee, D. Soudry, and N. Srebro (2018). "Characterizing Implicit Bias in Terms of Optimization Geometry". In: *ICML*.
Hahn, M., D. Jurafsky, and R. Futrell (2021). "Sensitivity as a Complexity Measure for Sequence Classification Tasks". In: *Transactions of the Association for Computational Linguistics*.
Jastrzebski, S., M. Szymczak, S. Fort, et al. (2020). "The Break-Even Point on Optimization Trajectories of Deep Neural Networks". In: *ICLR*.
Jiang, Y., B. Neyshabur, H. Mobahi, et al. (2020). "Fantastic Generalization Measures and Where to Find Them". In: *ICLR*.
Jordan, K., Y. Jin, V. Boza, et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*. Blog post. URL: https://kellerjordan.github.io/posts/muon/.
Kalra, D. S. and M. Barkeshli (2024). "Why Warmup the Learning Rate? Underlying Mechanisms and Improvements". In: *NeurIPS*.
Karpathy, A. (2022). *nanoGPT*. Software. URL: https://github.com/karpathy/nanoGPT.
Keskar, N. S. and R. Socher (2017). *Improving Generalization Performance by Switching from Adam to SGD*. Preprint.
Krizhevsky, A. (2009). *Learning Multiple Layers of Features from Tiny Images*. Tech. rep. University of Toronto.
Liu, J., J. Su, X. Yao, et al. (2025). *Muon is Scalable for LLM Training*. Preprint.
Moosavi-Dezfooli, S.-M., A. Fawzi, J. Uesato, and P. Frossard (2019). "Robustness via Curvature Regularization, and Vice Versa". In: *CVPR*.
Nagarajan, V. and J. Z. Kolter (2019). *Generalization in Deep Networks: The Role of Distance from Initialization*. NeurIPS 2017 Workshop on Deep Learning: Bridging Theory and Practice.
Neyshabur, B., R. Tomioka, and N. Srebro (2015). "Norm-Based Capacity Control in Neural Networks". In: *COLT*.
Novak, R., Y. Bahri, D. A. Abolafia, et al. (2018). "Sensitivity and Generalization in Neural Networks: an Empirical Study". In: *ICLR*.
Sanyal, A., P. H. S. Torr, and P. K. Dokania (2020). "Stable Rank Normalization for Improved Generalization in Neural Networks and GANs". In: *ICLR*.
Smith, S. L., B. Dherin, D. G. T. Barrett, and S. De (2021). "On the Origin of Implicit Regularization in Stochastic Gradient Descent". In: *ICLR*.
Soudry, D., E. Hoffer, M. S. Nacson, et al. (2018). "The Implicit Bias of Gradient Descent on Separable Data". In: *JMLR*.
Srinivas, S., K. Matoba, H. Lakkaraju, and F. Fleuret (2022). "Efficient Training of Low-Curvature Neural Networks". In: *NeurIPS*.
Vasudeva, B., J. W. Lee, V. Sharan, and M. Soltanolkotabi (2025). "The Rich and the Simple: On the Implicit Bias of Adam and SGD". In: *NeurIPS*.
Wilson, A. C., R. Roelofs, M. Stern, et al. (2017). "The Marginal Value of Adaptive Gradient Methods in Machine Learning". In: *NeurIPS*.
Xie, S. and Z. Li (2024). "Implicit Bias of AdamW: ℓ_∞ Norm Constrained Optimization". In: *ICML*.
Zhang, C., D. Zou, and Y. Cao (2024a). "The Implicit Bias of Adam on Separable Data". In: *NeurIPS*.
Zhang, Y., C. Chen, T. Ding, et al. (2024b). "Why Transformers Need Adam: A Hessian Perspective". In: *NeurIPS*.
Zhao, R., D. Morwani, D. Brandfonbrener, et al. (2025). "Deconstructing What Makes a Good Optimizer for Language Models". In: *ICLR*.

A Hyperparameters

Table 3: Optimizer hyperparameters, shared by all language-model settings unless noted. The Shakespeare control matches Muon's auxiliary AdamW to AdamW: $\beta = (0.9, 0.99)$, weight decay 0.1, learning rate $3 \cdot 10^{-4}$.

	SGD	AdamW	Muon
Update	heavy ball, $\mu = 0.9$, not Nesterov	$\beta = (0.9, 0.95)$; Shakespeare $(0.9, 0.99)$; $\epsilon = 10^{-8}$	Nesterov momentum 0.95; 5 Newton–Schulz steps (bfloat16); scale $\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Parameters	all	all	matrices in transformer blocks; auxiliary AdamW ($\beta = (0.9, 0.95)$, $\epsilon = 10^{-8}$) on embeddings, head, gains
Weight decay	none	0.1 on tensors with ≥ 2 dims, none on gains	none (Muon and auxiliary AdamW)
Clipping	none	none	none

Table 4: Learning-rate sweeps (data order 0, five rates per optimizer). Rates used in the main runs in bold. Selection: final validation loss for TinyStories, minimum validation loss over training for Shakespeare. The auxiliary AdamW of Muon keeps the learning rate of Table 1 throughout the sweep. SGD (†: best on data order 0) uses a lower rate, since its best rate diverged or blew up on other data orders: 0.2 on Shakespeare (orders 1 and 2), 0.1 on TinyStories-10M (orders 1–3) and 0.3 on TinyStories-30M (orders 2–4).

Setting	SGD	AdamW	Muon
Shakespeare-char	**0.05**, 0.1, 0.15, 0.2†, 0.25	$\mathbf{3 \cdot 10^{-4}}$, 10^{-3}, $2 \cdot 10^{-3}$, $4 \cdot 10^{-3}$, $8 \cdot 10^{-3}$	0.005, 0.01, 0.02, **0.04**, 0.08
TinyStories-10M	**0.03**, 0.1†, 0.3, 1, 3	$3 \cdot 10^{-4}$, 10^{-3}, $\mathbf{2 \cdot 10^{-3}}$, $4 \cdot 10^{-3}$, $8 \cdot 10^{-3}$	0.005, **0.01**, 0.02, 0.04, 0.08
TinyStories-30M	0.03, **0.1**, 0.3†, 1, 3	$3 \cdot 10^{-4}$, 10^{-3}, $\mathbf{2 \cdot 10^{-3}}$, $4 \cdot 10^{-3}$, $8 \cdot 10^{-3}$	0.005, **0.01**, 0.02, 0.04, 0.08

Table 5: Fork steps of the swaps (all six ordered pairs, plus each optimizer into itself with data order 1).

Setting	Fork steps
Shakespeare-char	25, 50, 100, 200, 300, 450, 600, 800, 1000
TinyStories-10M	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
TinyStories-30M	100, 200, 400, 800, 1400, 2000, 3000, 5000, 8000

Break-even runs: plain SGD (momentum 0), constant learning rate, no warmup. CIFAR-10: 100 epochs, i.e. 39,000 steps at batch 128, 166,600 at batch 30 and 16,600 at batch 300; data orders 0 and 1. Shakespeare and TinyStories-10M: 1,500 and 10,000 steps at their training batch size, learning rates 0.03, 0.1 and 0.3; data orders 0, 1 and 2.

B Measurement details

A grid "every $N + K$ log" is $\{0, N, 2N, \dots\} \cup \{\text{round}(T^u) : u \in K \text{ evenly spaced values in } [0, 1]\} \cup \{T\}$, i.e. the union of a linear grid and K log-spaced steps between 1 and T.

Sharpness. Lanczos runs for 20 iterations with full reorthogonalization (two Gram–Schmidt passes per iteration) from a fixed Gaussian start, and returns the top Ritz pair. Hessian-vector products use double backpropagation through one gradient graph, with the exact (math) attention kernel, in float32 without autocast, on the model before the update of the measured step.

Split. $\tau = 0.02$, $m_\times = 0.5$, $m_+ = 0$, $k = 3$, 1000 bootstrap resamples. The training loss of a run is its loss on the training probe at its prediction steps.

Mode connectivity. Barrier on the 64 sequences of the training probe, at 11 evenly spaced values of α, between final checkpoints.

Prediction trajectories. 128 evenly spaced probe positions; top 50 principal components of the centered probability vectors, then UMAP.

Table 6: Logging grids, probe sizes and measurement settings of the main language-model runs (forks inherit their parent's) and of the CIFAR-10 runs at batch 128. Fixed sets of examples are drawn once, from random streams separate from the training batches, and are the same for all runs of a setting.

	Shakespeare-char	TinyStories-10M	TinyStories-30M	CIFAR-10 (batch 128)
Prediction steps $\mathcal{S}$	every 50 + 40 log	every 250 + 30 log	every 500 + 30 log	every 390 + 40 log
Checkpoints	every 250 + 12 log	every 1000 + 12 log	every 2000 + 12 log	every 3900 + 12 log
Measurement steps $\mathcal{M}$	every 50 + 30 log	every 500 + 30 log	every 1000 + 30 log	every 390 + 30 log
Validation probe (N)	$64 \times 256 = 16{,}384$	$128 \times 16 = 2{,}048$	$128 \times 16 = 2{,}048$	10,000 images
Training probe	64 seq. (16,384 tok.)	64 seq. (16,384 tok.)	64 seq. (16,384 tok.)	2,000 images
Hessian batch $\mathcal{H}$	32 seq. (8,192 tok.)	32 seq. (8,192 tok.)	32 seq. (8,192 tok.)	2,500 images
Lanczos iterations	20	20	20	20
Batch sharpness	8×64 seq.	8×128 seq.	8×128 seq.	8×128 images
$\hat{K}$ (break-even runs)	25×64 seq.	25×128 seq.	–	25×128 images

B.1 Function metrics

The metrics are computed on 1024 fixed validation examples (sequences or test images), drawn separately from the probe and shared by all runs; the input-derivative metrics use the first 32.

Input derivatives. Let $f(x)$ be the loss at the true labels as a function of the input x. For the language models $x \in \mathbb{R}^{64\times d}$ is the token-embedding matrix of the first 64 tokens of a sequence (position embeddings are added inside the model) and f is the mean next-token cross-entropy over these 64 positions; for the CNN x is the normalized image and f its cross-entropy. With $g = \nabla f(x)$, $H = \nabla^2 f(x)$ and $T[u,u,\cdot] = \nabla_x\big(u^\top H(x)u\big)$, and $z_1, \dots, z_8$ standard Gaussian probes:

- gradient norm $\|g\|$;
- Jacobian norm $\|J\|_F$ of the softmax output with respect to x (Novak et al., 2018): exact for the CNN (one vector-Jacobian product per class), and estimated as $\big(\frac{1}{8}\sum_j \|J^\top z_j\|^2\big)^{1/2}$ over the $64 \times V$ outputs for the language models;
- Hessian norm $\|H\|_F \approx \big(\frac{1}{8}\sum_j \|Hz_j\|^2\big)^{1/2}$;
- top input-Hessian eigenvalue $\lambda_x = v^\top Hv$, with v from 20 power iterations (the eigenvalue of largest magnitude) (Moosavi-Dezfooli et al., 2019);
- level-set curvature $\big(\frac{1}{8}\sum_j \|PHPz_j\|^2\big)^{1/2}/\|g\| \approx \|PHP\|_F/\|g\|$, with $P = I - \hat{g}\hat{g}^\top$ the projection orthogonal to the gradient, a projected variant of the normalized curvature of Srinivas et al. (2022);
- third order: the norm $\|T\|_\sigma = \max_{\|u\|=1} T[u,u,u]$, by shifted symmetric power iteration $u \leftarrow (T[u,u,\cdot] + \alpha u)/\|T[u,u,\cdot] + \alpha u\|$ with $\alpha = \|T[u_0,u_0,\cdot]\|$, 20 iterations, best of 3 random starts (as $T[u,u,u]$ is odd in u, this is also $\max|T[u,u,u]|$); the third derivative along the sharpest direction $T[v,v,v]$, with the sign of v fixed by $v^\top g \geq 0$; and the curvature length $|\lambda_x|/\|T[v,v,\cdot]\|$, the distance over which, to first order, λ_x changes by its own size (for a simple eigenvalue $\nabla_x \lambda_x = T[v,v,\cdot]$).

Each is reported as mean and standard deviation over the 32 points.

Weight norms. Over all parameters θ: ℓ_0 (entries with $|\theta_i| > 10^{-8}$), ℓ_1, ℓ_2, ℓ_3, ℓ_∞. For each weight tensor with at least two dimensions, reshaped to a matrix W (output $\times$ input): the spectral norm $\|W\|_2$, nuclear norm $\|W\|_*$ and stable rank $\|W\|_F^2/\|W\|_2^2$ (Sanyal et al., 2020). Distance from initialization $\|\theta - \theta_0\|_2$, also relative to $\|\theta_0\|_2$ (Nagarajan and Kolter, 2019). The log spectral complexity $\log\big(\prod_i \|W_i\|_2 \cdot (\sum_i (\|(W_i - W_i^0)^\top\|_{2,1}/\|W_i\|_2)^{2/3})^{3/2}\big)$ of Bartlett et al. (2017), with the initialization as reference matrices. For the CNN only, the path norm as computed by Jiang et al. (2020) (after Neyshabur et al., 2015): the sum of the outputs on an all-ones input of the network with every parameter squared and identity activations.

Accuracy and robustness. Accuracy on the 1024 validation examples (for the language models, next- token accuracy over all positions). For the CNN, accuracy under Gaussian input noise of standard deviation 0.05, 0.1 and 0.2 (in normalized pixel units) (Ford et al., 2019). For the language models, a one-token substitution sensitivity after Hahn et al. (2021): for 32 validation contexts of 64 tokens, we replace each position in turn by a uniformly random different token and count how many of the 64 substitutions change the argmax prediction at the last position; we report this count as a fraction of 64, averaged over contexts. All derivatives are taken by automatic differentiation in float32, with the exact (math) attention kernel.

C Compute

All runs used one GPU each on the MIT ORCD cluster [TODO: GPU types]. [TODO: GPU-hours per suite and total.]

Table 7: Runs per suite and compute.

Suite	Runs	GPU-hours
Learning-rate sweeps (3 settings)	3×15	??
Main runs (3 settings)	3×15	??
Forks (3 settings)	3×81	??
Break-even, Shakespeare and TinyStories-10M	2×9	??
Break-even, CIFAR-10	10	??
Shakespeare control (matched Muon)	6	??
Offline function metrics and mode connectivity	–	??
Total	367	??

D Additional figures and tables

Table 8: Edge of stability in the language models: end of warmup w; measured steps at the edge of stability ($\mathcal{Q} \geq 0.9\,\Theta$; $\lambda_{\max}$ for SGD, preconditioned sharpness for AdamW) of the $d = 0$ run, from first to last; onset s_{EoS} over the 5 data orders; fraction of measured steps at the edge of stability over the 5 data orders; first exceedance t^*_X of the $d = 0$ run against the other two optimizers (in the order SGD, AdamW, Muon); and its persistent split, if any.

Setting	Run	w	at EoS ($d = 0$)	s_{EoS} (all d)	fraction of $\mathcal{M}$	t^*_X	$t^*_{\mathrm{p},X}$
Shakespeare	SGD	100	21–120	21	2–14%	1 / 1	191 (vs Muon)
	AdamW	100	1–500	1	11–16%	1 / 1	–
	Muon	100	–	–	–	1 / 2	14 (vs SGD)
TinyStories-10M	SGD	200	62–1500	45–500	6–12%	3 / 1	–
	AdamW	200	1–4500	1	50–65%	1 / 1	–
	Muon	200	–	–	–	1 / 7	–
TinyStories-30M	SGD	400	60–658	60–467	10–16%	2 / 1	–
	AdamW	400	1–14000	1	40–69%	1 / 1	–
	Muon	400	–	–	–	1 / 8	–

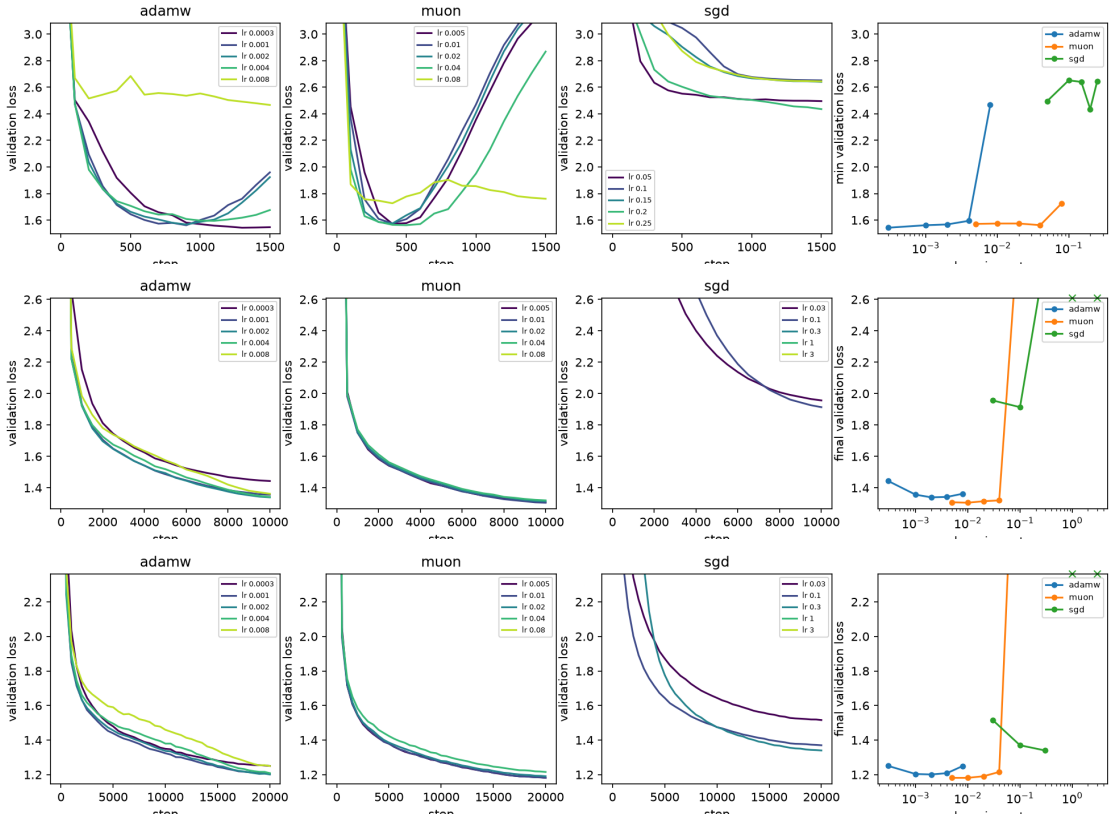


Figure 8: Learning-rate sweeps on Shakespeare (top), TinyStories-10M and TinyStories-30M (bottom), data order 0. One panel per optimizer with the validation loss of each rate against step, and a last panel with the minimum (Shakespeare) or final (TinyStories) validation loss against the learning rate; × marks runs that diverged. Curves above the plotted range are cut.

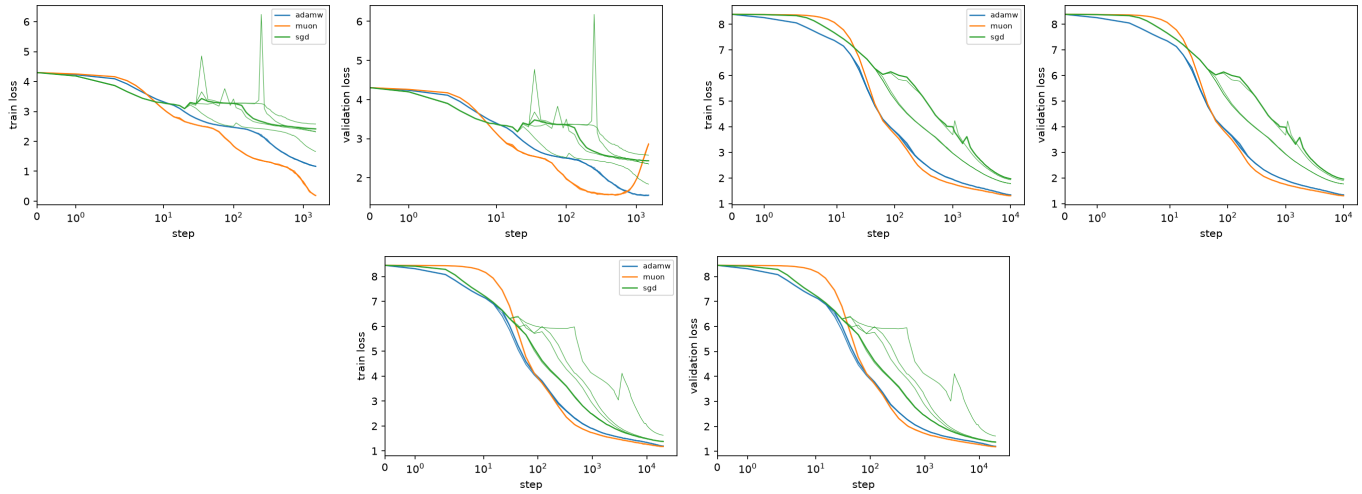


Figure 9: Training-probe and validation loss of the main runs against step: Shakespeare (top left), TinyStories-10M (top right), TinyStories-30M (bottom). Data order 0 thick, orders 1–4 thin.

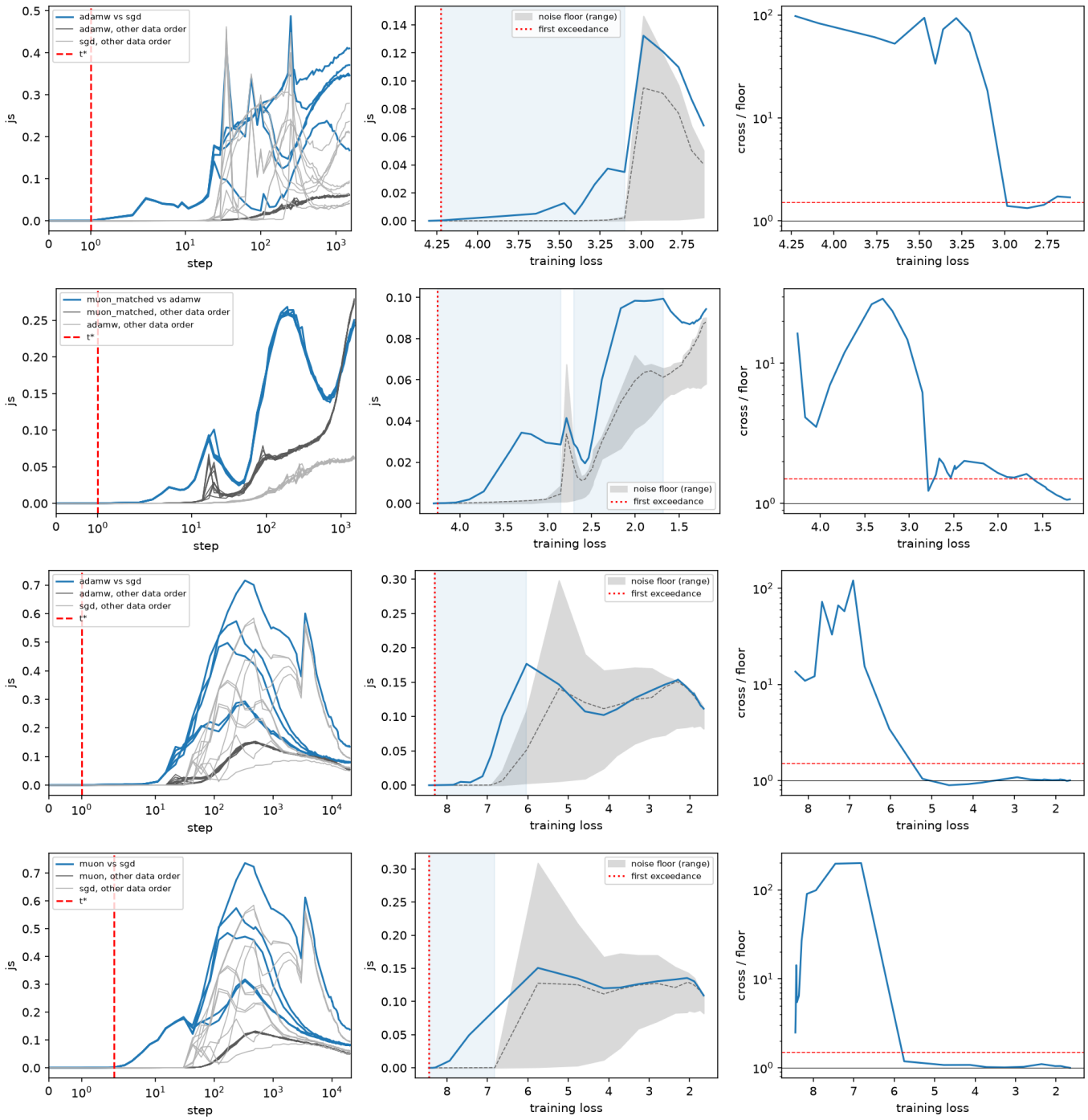


Figure 10: Divergence figures as in Fig. 1. Rows: Shakespeare AdamW–SGD; Shakespeare matched Muon–AdamW (control); TinyStories-30M AdamW–SGD; TinyStories-30M Muon–SGD.

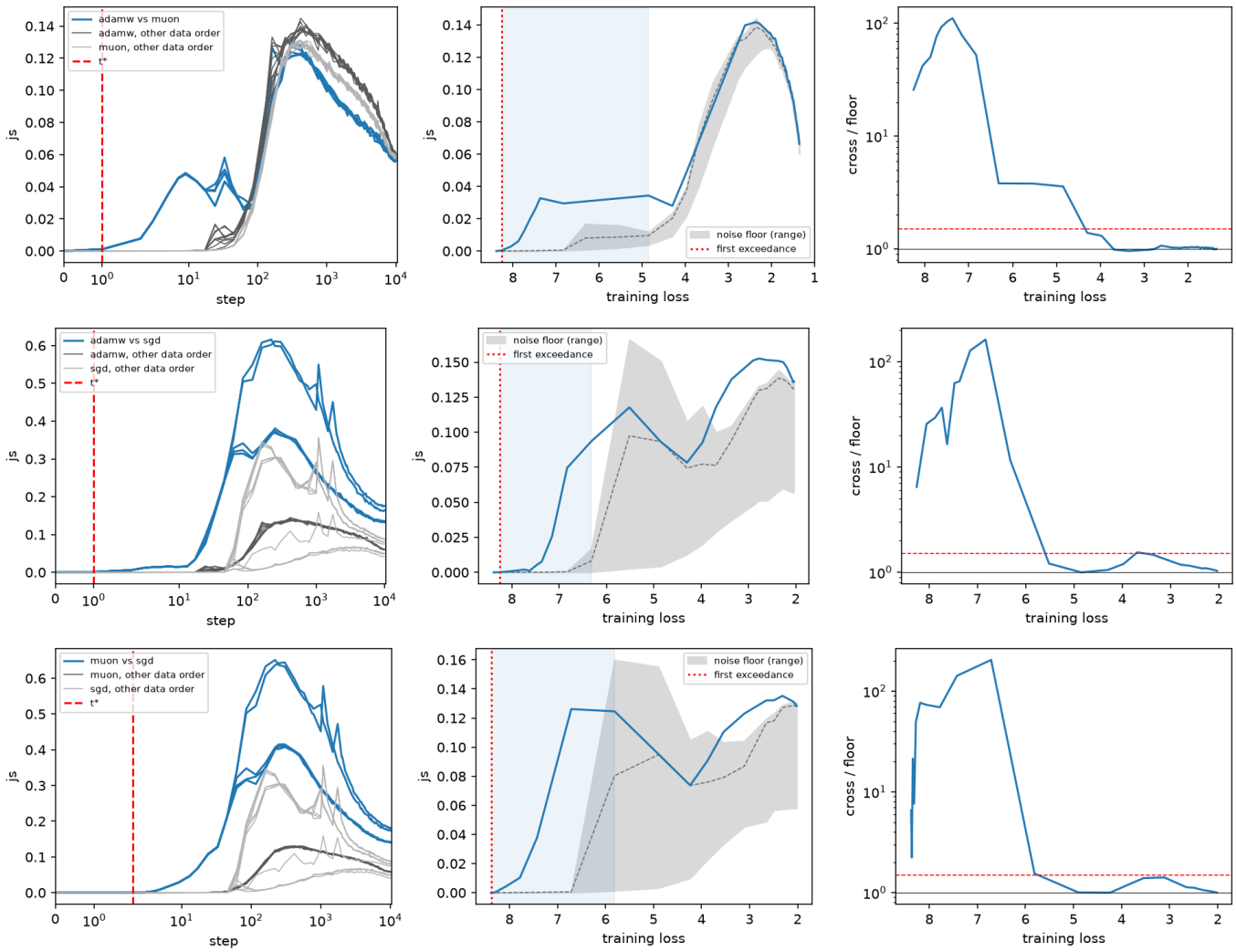


Figure 11: Divergence figures as in Fig. 1 for TinyStories-10M. Rows: AdamW–Muon, AdamW–SGD, Muon–SGD.

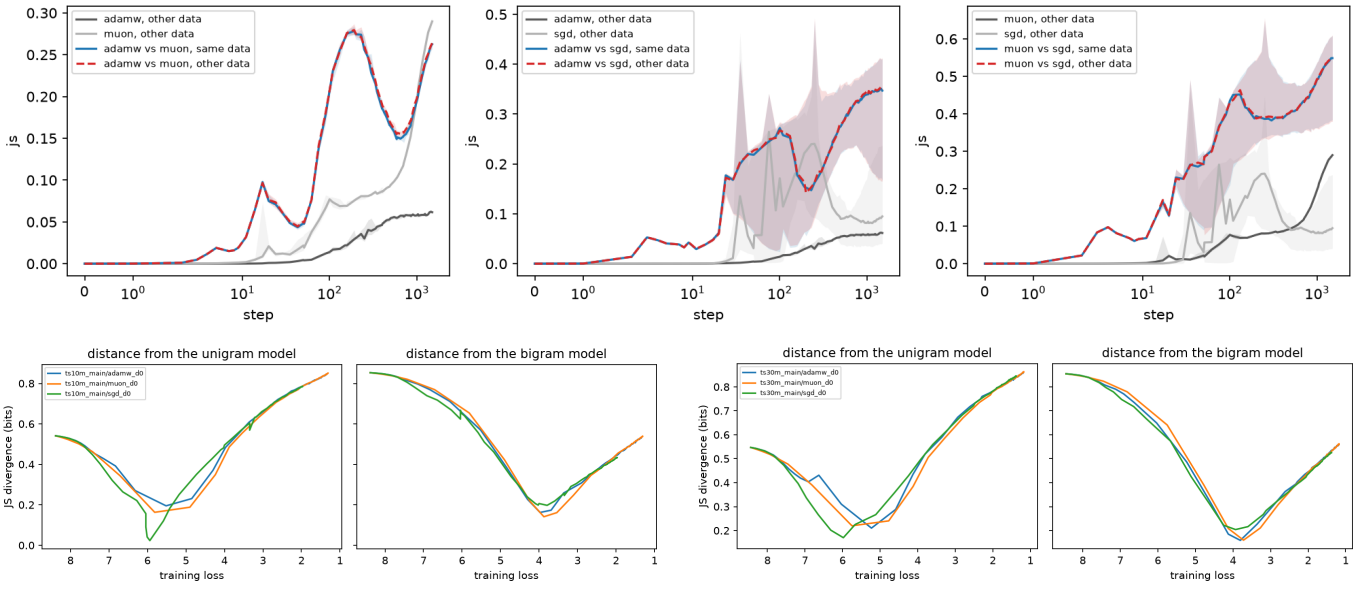


Figure 12: Top: optimizer against data order at equal steps on Shakespeare, as in Fig. 2 (top), for the three pairs. Bottom: distance of the $d = 0$ runs from the unigram and bigram models on TinyStories-10M (left) and TinyStories-30M (right), as in Fig. 2 (bottom).

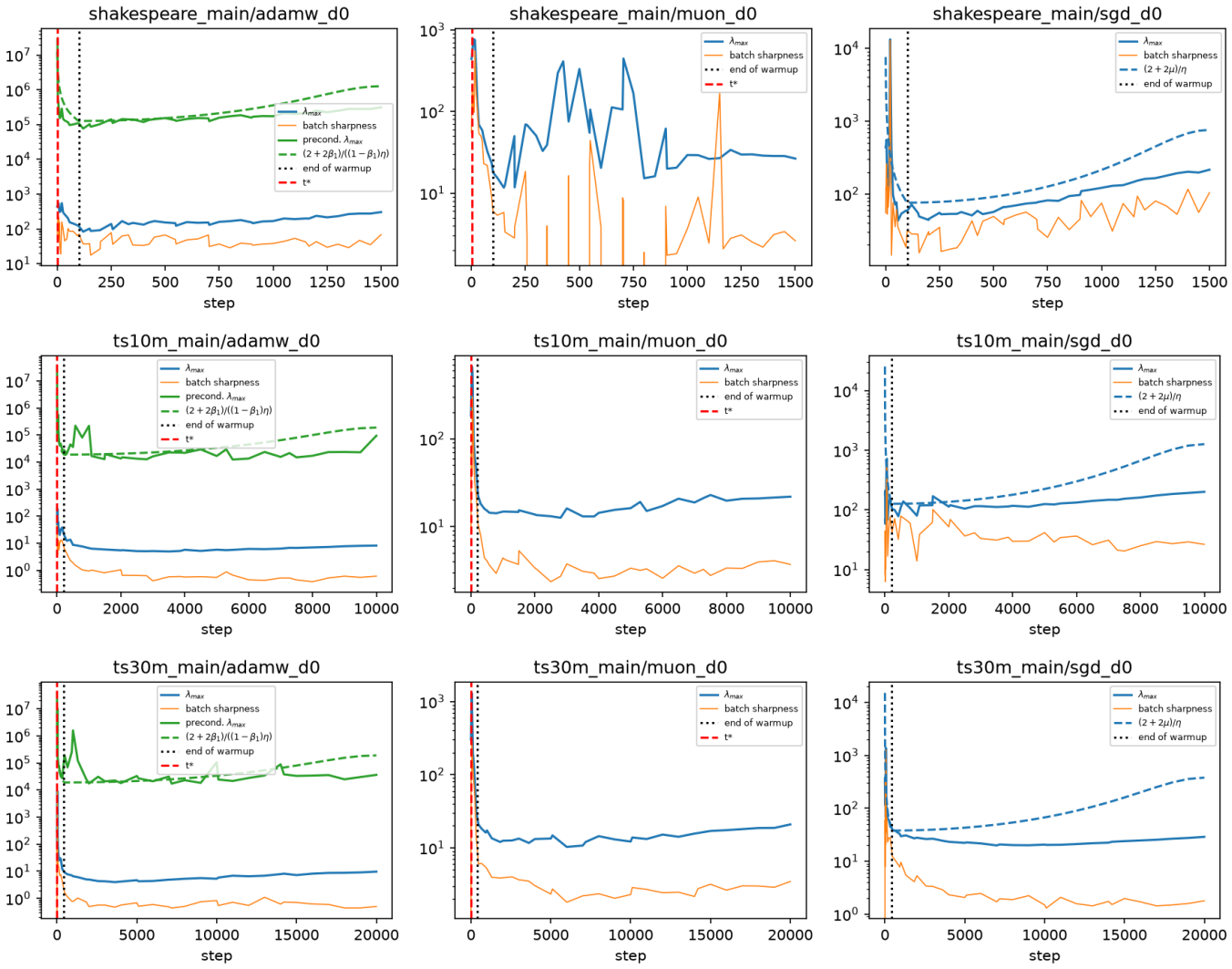


Figure 13: Sharpness of the $d = 0$ runs on Shakespeare (top), TinyStories-10M and TinyStories-30M (bottom), log scale: $\lambda_{\max}$, batch sharpness, for AdamW the preconditioned sharpness, and the thresholds $(2 + 2\mu)/\eta_s$ (SGD) and $(2 + 2\beta_1)/((1 - \beta_1)\eta_s)$ (AdamW, dashed); end of warmup dotted, first exceedance t^*_X of the AdamW–Muon pair red dashed.

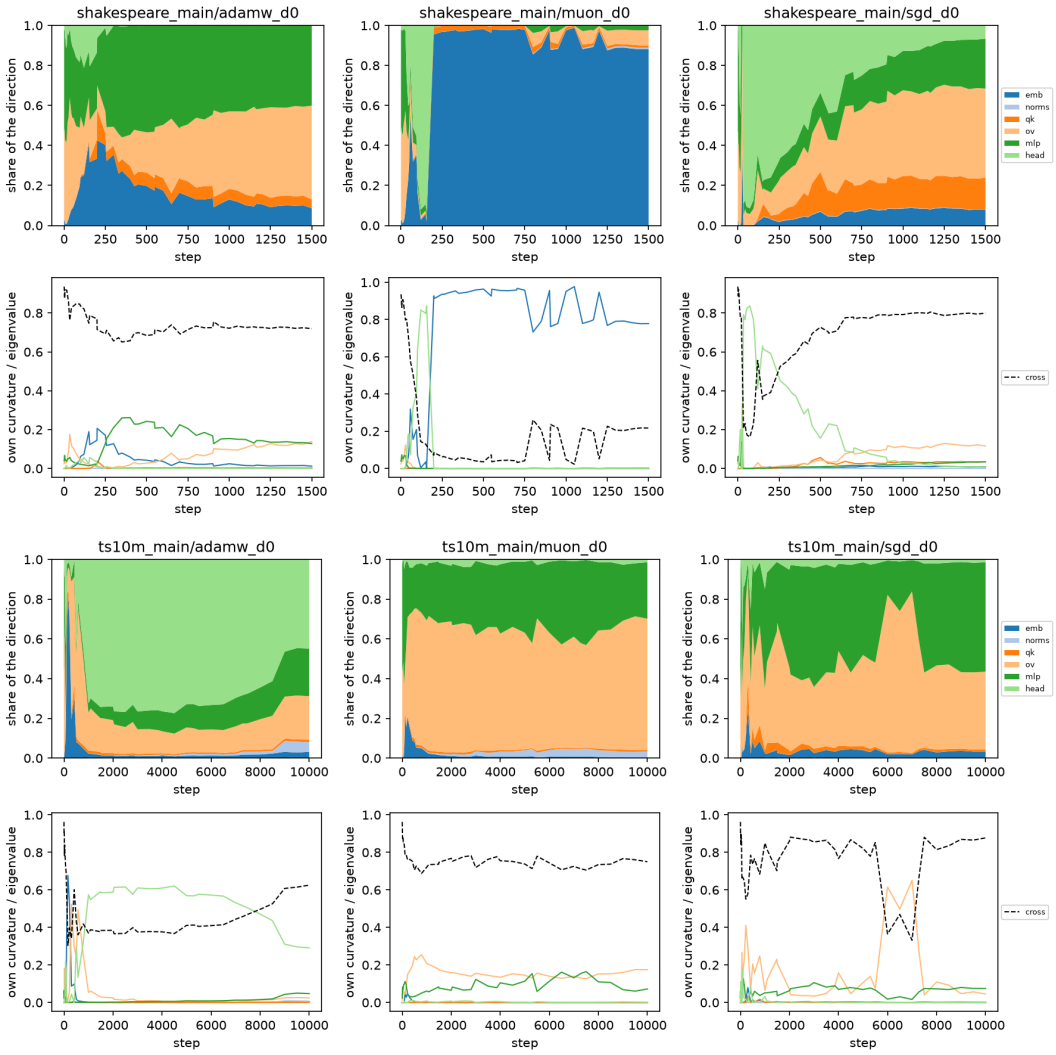


Figure 14: Anatomy of the top Hessian eigenvector as in Fig. 5, on Shakespeare (top two rows) and TinyStories-10M (bottom two rows).

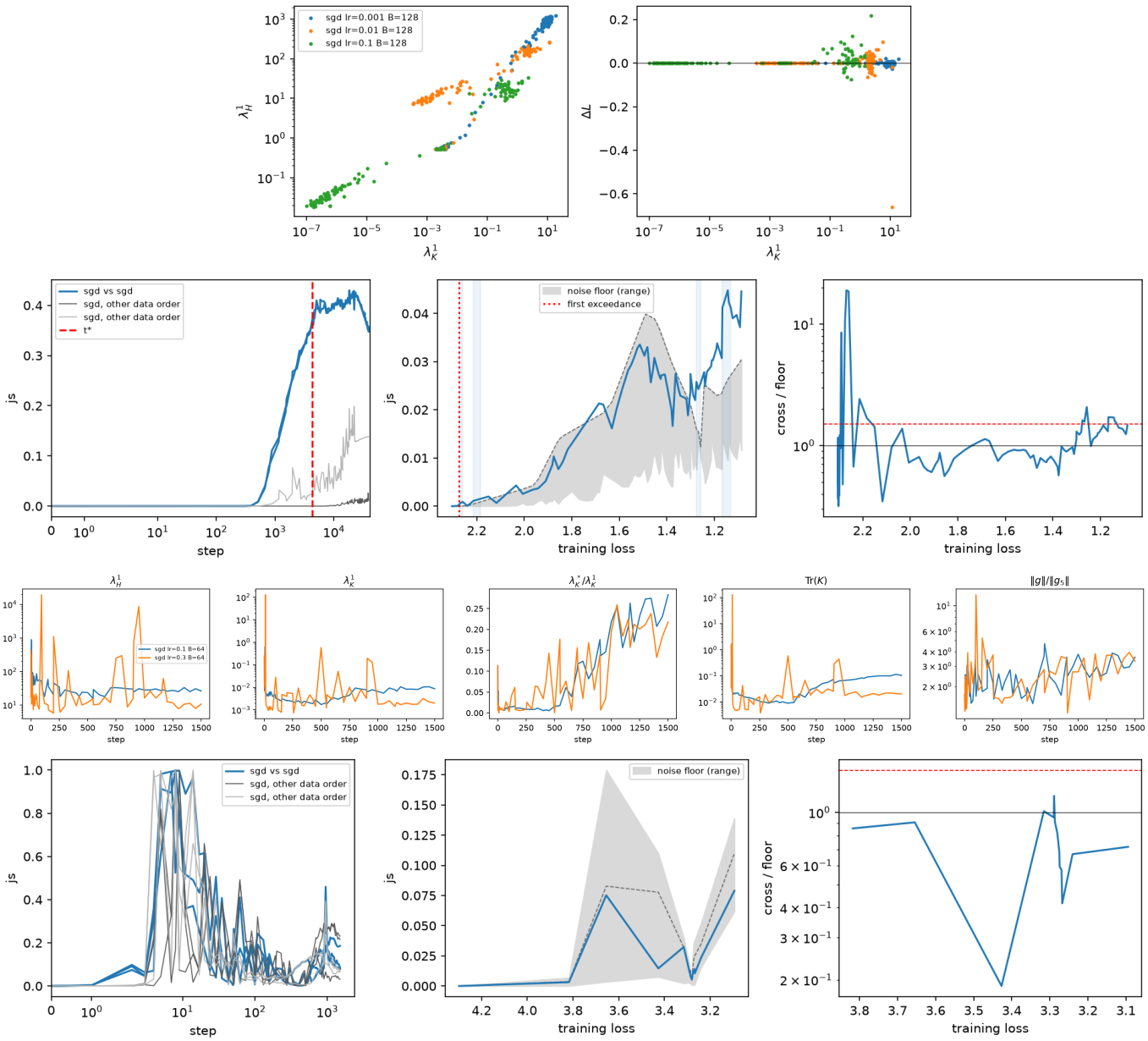


Figure 15: Break-even analysis, continued. First row: CIFAR-10, λ_H^1 (left) and ΔL (right) against λ_K^1. Second row: CIFAR-10 divergence figure for $\eta = 0.001$ against 0.01. Third and fourth rows: Shakespeare with plain SGD at $\eta \in \{0.1, 0.3\}$, the quantities of Fig. 6 (top) and the divergence figure.